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Module 7:

Periodic quantities

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1 Introduction

In the preceding modules, the fundamental physical principles and the basic components have been discussed. Furthermore, the essential technical behaviours in direct-current (DC) networks were clarified. In DC networks, electrical circuits are supplied with direct current, whereas alternating-current (AC) networks operate with alternating current. DC flows continuously in one direction, while AC periodically reverses its direction. The regularity of this reversal follows a specific frequency. For example, an ideal electrical resistor is not frequency-dependent. In contrast, components such as capacitors and inductors exhibit different complex impedances at different frequencies. To analyse these periodic quantities, the following topics are addressed:

- Pointer diagram
- Complex AC circuit analysis
- Effective value
- Apparent power, active power and reactive power
- Three-phase current
- Co-system, counter-system and zero system

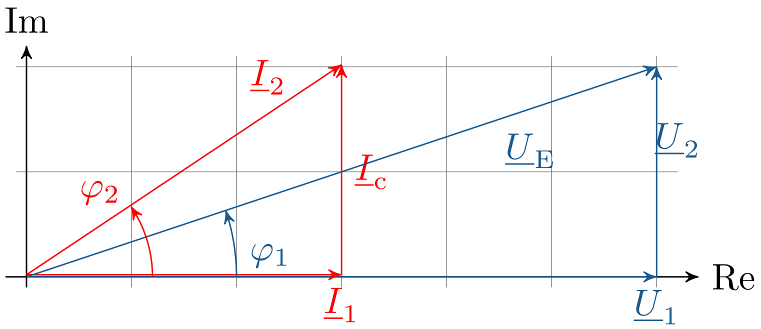


Figure 1.1: **Pointer diagram.** The quantities voltage and current are represented as vectors.

2 Fundamentals of Complex Numbers

The phasor diagrams used in alternating current technology provide a quick overview of the magnitude and direction of the voltage and current. The underlying concepts of the complex number plane and the structure of phasor diagrams are explained in more detail below and illustrated with an example.

Learning objectives: Complex numbers

The Students can

- handle numbers in the complex plane.
- Plot phasor diagrams of complex numbers.
- calculate complex numbers.

2.1 Complex number plane

In order to understand the structure of pointer diagrams, a basic understanding of the complex number system is necessary. For elementary arithmetic operations, natural numbers including zero and rational numbers are sufficient. Rational numbers can be represented as finite or periodic decimal numbers. Irrational numbers, on the other hand, can be represented as decimal numbers that have an infinite number of digits and are not periodic. Real numbers are composed of rational numbers and irrational numbers. However, it is not possible to take the square root of a negative number in the real number system. For this purpose, the complex number system is introduced. In the complex number system, the real number space is extended by the imaginary unit j. Thus, in the complex number system, taking the square root of -1 results in the imaginary unit j.

$$\mathrm{j} = \sqrt{-1} \tag{2.1}$$

Squaring the imaginary unit again gives -1.

$$\mathrm{j}^2 = -1 \tag{2.2}$$

A complex number $\underline{Z}$ describes a location in the complex plane. To uniquely define a location in a two-dimensional coordinate system, two coordinates are required. The two coordinates for describing a complex number $\underline{Z}$ are described in the complex plane as the real part and the imaginary part (see Equation 2.3). Complex numbers are usually indicated by an underscore, whereby the real part and the imaginary part represent real numbers.

$$\underline{Z} = \mathit{Realpart} + \mathrm{j} \cdot \mathit{Imaginary\ part} \tag{2.3}$$

$$\underline{Z} = \mathrm{Re}(\underline{Z}) + \mathrm{j} \cdot \mathrm{Im}(\underline{Z}) \tag{2.4}$$

In the complex plane, the real part is plotted on the abscissa and the imaginary part on the ordinate. The abbreviations Re = real part and Im = imaginary part are used. The coordinate system of a complex plane is explained in Figure 2.1. The location of the complex number can be represented by direction arrows, also known as vectors or pointers. The representation of the complex number in Cartesian coordinates is achieved by decomposing it into the real part of the complex number

Re($\underline{Z}$) and the imaginary part of the complex number Im($\underline{Z}$) combined with the imaginary unit j. Complex numbers can also be represented in polar coordinates. This is done by the magnitude $|\underline{Z}|$ of the complex number $\underline{Z}$ and by the angle φ that the pointer of the complex number encloses with the real axis.

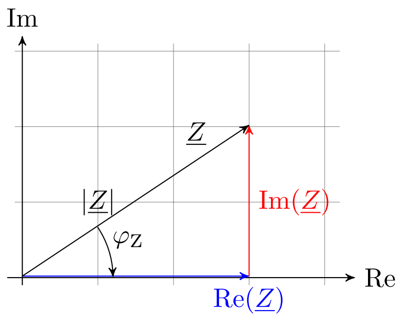


Figure 2.1: **Pointer diagram of a complex number.** A complex number $\underline{Z}$, which can be represented by $Re(\underline{Z})$ and $Im(\underline{Z})$ in Cartesian coordinates and by the magnitude of the complex number $|\underline{Z}|$ and the corresponding angle φ in polar coordinates.

The two forms of representation can be transformed into each other. Euler's formula is helpful here. Euler's formula shows that the ordinate value and the abscissa value of the unit circle can be calculated using the trigonometric functions cosine and sine. Euler's formula is represented in equation 2.5.

$$\mathrm{e}^{\mathrm{j}\cdot\chi} = \cos(\chi) + \mathrm{j}\cdot\sin(\chi) \tag{2.5}$$

Using Euler's formula, the polar coordinates can be transformed into Cartesian coordinates by adding the absolute value of the complex number. This is explained by equation 2.6.

$$\underline{Z} = |\underline{Z}|\cdot\cos(\varphi) + \mathrm{j}\cdot|\underline{Z}|\cdot\sin(\varphi) = \mathrm{Re}(\underline{Z}) + \mathrm{j}\cdot\mathrm{Im}(\underline{Z}) \tag{2.6}$$

Using Pythagoras' theorem, the real and imaginary parts of the complex number can be used to determine the magnitude of the complex number for the polar coordinates. The corresponding angle is obtained from the arctan with the ratio of the imaginary part to the real part in the argument. With this information, a complex number such as in equation 2.7 can be transformed into polar coordinates.

$$\underline{Z} = \sqrt{(\mathrm{Re}(\underline{Z}))^2 + (\mathrm{Im}(\underline{Z}))^2}\cdot\mathrm{e}^{\mathrm{j}\cdot\arctan\left(\frac{\mathrm{Im}(\underline{Z})}{\mathrm{Re}(\underline{Z})}\right)} = |\underline{Z}|\cdot\mathrm{e}^{\mathrm{j}\cdot\varphi} \tag{2.7}$$

Key point: Representations of complex numbers

Cartesian representation:

$$\underline{Z} = \mathrm{Re}(\underline{Z}) + j\cdot\mathrm{Im}(\underline{Z})$$

Polar form:

$$\underline{Z} = |\underline{Z}|\cdot e^{j\cdot\varphi}$$

Trigonometric representation:

$$\underline{Z} = |\underline{Z}|(\cos(\varphi) + j\cdot\sin(\varphi))$$

Conjugation is one of the operations on complex numbers. When conjugating a complex number, the j is replaced by -j (negation of the imaginary part). As shown in Equation 2.8, conjugation is indicated by an asterisk.

$$\underline{Z}^* = (\mathrm{Re}(\underline{Z}) + \mathrm{j} \cdot \mathrm{Im}(\underline{Z}))^* = \mathrm{Re}(\underline{Z}) - \mathrm{j} \cdot \mathrm{Im}(\underline{Z}) \tag{2.8}$$

The magnitude of a complex number can be determined by multiplying the complex number by its conjugate (see Equation 2.9). Instead of the magnitude, which contains a root, the square of the magnitude is also used. It also represents a measure of the distance of the number from the origin, but is easier to calculate.

$$|\underline{Z}| = \sqrt{\underline{Z} \cdot \underline{Z}^*} \rightarrow |\underline{Z}|^2 = \underline{Z} \cdot \underline{Z}^* \tag{2.9}$$

When adding and subtracting complex numbers, it is advisable to first convert them into Cartesian coordinates. This allows the real part and the imaginary part of the complex number to be calculated separately.

$$\underline{Z}_1 + \underline{Z}_2 = \mathrm{Re}(\underline{Z}_1) + \mathrm{Re}(\underline{Z}_2) + j \cdot (\mathrm{Im}(\underline{Z}_1) + \mathrm{Im}(\underline{Z}_2)) \tag{2.10}$$
$$\underline{Z}_1 - \underline{Z}_2 = \mathrm{Re}(\underline{Z}_1) - \mathrm{Re}(\underline{Z}_2) - j \cdot (\mathrm{Im}(\underline{Z}_1) - \mathrm{Im}(\underline{Z}_2)) \tag{2.11}$$

The representation in polar coordinates is recommended for calculating complex numbers by multiplication and division. The multiplication of complex numbers is composed of the product of the magnitudes and the sum of the respective angles (see Equation 2.12). In division, the magnitudes are divided and the angles are subtracted from each other (see Equation 2.13).

$$\underline{Z}_1 \cdot \underline{Z}_2 = |\underline{Z}_1| \cdot |\underline{Z}_2| \cdot e^{j \cdot (\varphi_1 + \varphi_2)} \tag{2.12}$$

$$\frac{\underline{Z}_1}{\underline{Z}_2} = \frac{|\underline{Z}_1|}{|\underline{Z}_2|} \cdot e^{j \cdot (\varphi_1 - \varphi_2)} \tag{2.13}$$

Key point: Basic arithmetic operations with complex numbers

For **addition and subtraction**, the **Cartesian representation** is generally used for complex numbers. For **multiplication and division** of complex numbers, the **polar form** is chosen.

Since the division is initially only defined for real numbers other than zero, the complex number in the denominator is extended so that it becomes real.

$$\frac{\underline{Z}_1}{\underline{Z}_2} = \frac{\underline{Z}_1}{\underline{Z}_2} \cdot \frac{\underline{Z}_2^*}{\underline{Z}_2^*} = \frac{\underline{Z}_1 \cdot \underline{Z}_2^*}{|\underline{Z}_2|^*} \tag{2.14}$$

Adding a complex number to its conjugate cancels out its imaginary part (see equation 2.15). Similarly, subtracting the real part cancels it out (see equation 2.16).

$$\mathrm{Re}(\underline{Z}) = \frac{\underline{Z} + \underline{Z}^*}{2} \tag{2.15}$$

$$\mathrm{Im}(\underline{Z}) = \frac{\underline{Z} - \underline{Z}^*}{2j} \tag{2.16}$$

2.2 Graphical representation and calculations with complex numbers

Complex numbers can also be solved graphically using pointer diagrams. If the complex numbers are represented in Cartesian form as recommended, the components of the real parts and the imaginary parts can be read directly and entered into a coordinate system. The complex numbers are entered, for example, in Figure 2.2. When adding complex numbers, either Z_1 is set to Z_2 or vice versa. This is because the commutative law still applies to addition. Here, the dashed arrows indicate the parallel shifts of Z_1 and Z_2. Both shifts end at the same coordinate. Here, the real part and the imaginary part can be read separately from the sum of the two complex numbers.

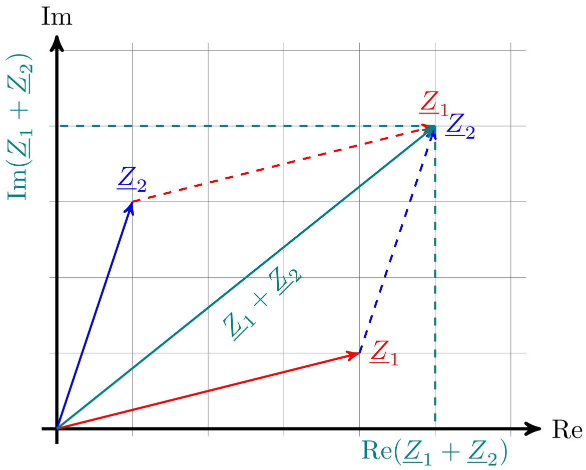


Figure 2.2: **Addition of complex numbers.** Graphical solution for adding two complex numbers in a phasor diagram.

The graphical solution for subtracting complex numbers is similar to that for addition. It should be noted that, as in other number systems, the commutative law does not apply when subtracting two complex numbers. If the complex number Z_2 is subtracted from Z_1, the direction of Z_1 changes. Here, the parallel shift occurs again, but only for Z_1. The resulting new vector can then be read again as the real part and the imaginary part.

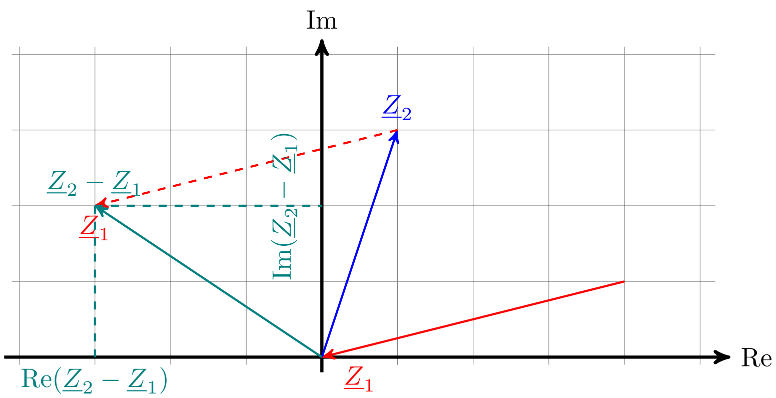


Abbildung 2.3: **Subtraction of complex numbers.** Graphical solution for subtracting two complex numbers in a phasor diagram.

The complex numbers $\underline{Z}_1$ and $\underline{Z}_2$ shown in Figure 2.4 are represented for multiplication in polar coordinates. Using the previously described calculation rules for multiplication of complex numbers, the values for the result $\underline{Z}_3$ can be determined. The multiplication of the pointer lengths gives the length of the product. The sum of the two complex numbers gives the angle of the new complex number.

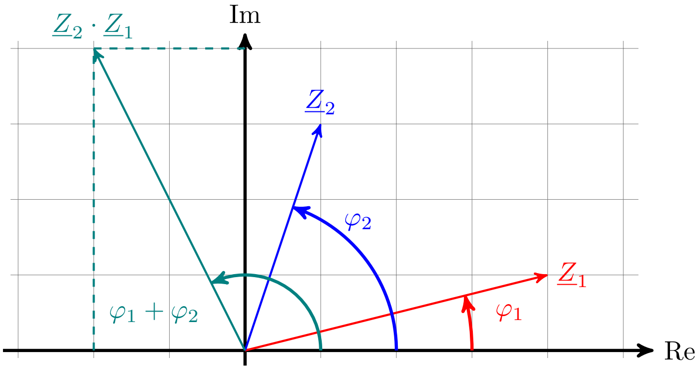


Abbildung 2.4: **Multiplication of complex numbers.** Graphical solution for multiplying two complex numbers in polar coordinates in a pointer diagram.

3 Pointer diagrams in alternating current technology

In alternating current technology, complex numbers are used, among other things, as pointers. In linear networks, sinusoidal and single-frequency source quantities only result in sinusoidal and single-frequency voltages and currents. In a steady state, the voltages and currents only differ in their amplitude and the associated phase. All signals have an identical frequency. Steady-state and harmonic oscillations can be represented by their time course in a line diagram or by complex numbers using pointers. Describing oscillations using complex numbers allows them to be represented by stationary pointers, whereby only their relative position to each other is expressed.

Learning objectives: Pointer diagrams

the students can

- understand the parameters of periodic alternating voltages.
- can handle the complex rotation indicators of the amplitudes.

3.1 Periodic alternating voltage

Figure 3.1 shows an alternating voltage signal with a peak value (amplitude value) of $\hat{U}$. The offset of the alternating voltage signal on the abscissa is expressed by the time t_0. The point at which the alternating voltage signal crosses the ordinate indicates the alternating voltage value U at time $t = 0$.

The alternating voltage signal is periodic in T. A period is defined for the specified alternating voltage signal by a complete cycle of a positive and negative half-wave. Thus, starting from t_0, after one period, the time $t_0 + T$ is reached, and after two periods, the time $t_{0+2\mathrm{T}}$ is reached.

By transferring the specified values to a circular diagram, the relationship between the alternating voltage and the complex representation can be explained. In this way, the alternating voltage magnitude $U(t = 0)$ at time 0 can be described by the length and orientation of the pointer. The peak value corresponds to the length of the pointer in the circular diagram. The angle φ_u describes the phase angle at time $t = 0$.

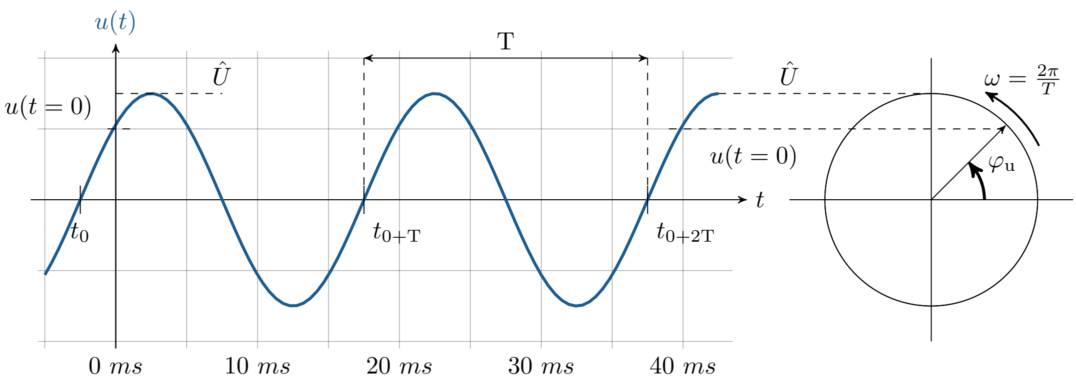


Figure 3.1: **Periodic alternating voltage over time with two periods.** The alternating voltage can be used to determine, for example, the amplitude value and the period duration.

The angular frequency ω can be specified using the period T or the frequency f. Furthermore, the angular frequency is also described as angular velocity. Using the relationship between the period and the frequency, the quantities can be converted into one another. The relationships explained are described in equations 3.1.

$$\omega = \frac{2\pi}{T} \quad \text{with} \quad f = \frac{1}{T} \quad \text{will be} \quad \omega = 2\pi f \tag{3.1}$$

Assuming that the cycle of a complete period at a frequency of $50Hz$ takes $20ms$ (see Figure 3.2). Applied to the degree measure and the wheel measure, this results in angle specifications for the cycle of a period. The angle after the complete period T corresponds to the rotation of a full circle. The rotation of a full circle corresponds to the phase shift t_0 can also be described in angular representation as the phase angle φ_{N}.

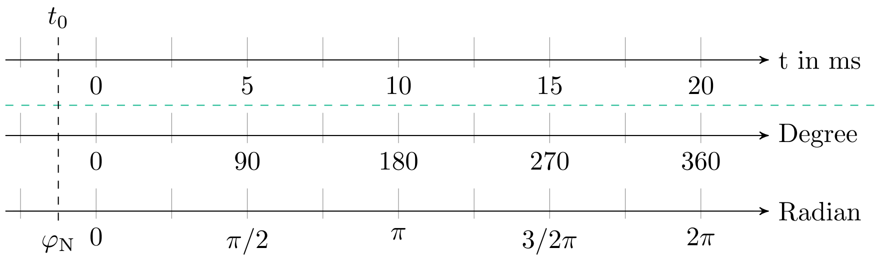


Abbildung 3.2: **Various measurements.** Comparison between degrees and radian measure: the circumference of a circle corresponds to 360^o or 2π.

Key point: Exchangeable sizes

An alternating voltage explains a regular polarity change with the amplitude value $\hat{U}$ and the **Periodendauer** T.

The time-dependent voltage value $u(t)$ of an alternating voltage depends on several influencing factors. These include the peak value and frequency of the alternating voltage as well as the corresponding phase angle. $u(t)$ can be determined from the ratio of the time t to the period duration.

$$u(t) = \hat{U} \cdot \sin(\omega t + \varphi_{\mathrm{N}}) = \hat{U} \cdot \sin(2\pi f t + \varphi_{\mathrm{N}}) = \hat{U} \cdot \sin(2\pi \frac{t}{T} + \varphi_N) \tag{3.2}$$

3.2 Complex rotating pointer of the amplitude

The complex voltage value and the complex current value can be represented as pointers. Here, the polar coordinate representation is used.

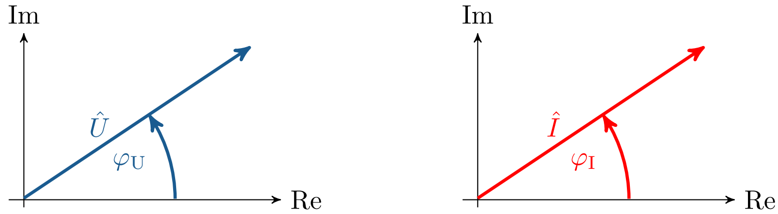


Figure 3.3: **Voltage and current indicators.** A voltage vector with associated phase angle and, equivalently, a current vector with phase angle

A harmonic alternating quantity can be represented as a pointer rotating at a constant angular velocity ω. The time function is the projection of the pointer onto the real axis. The rotating pointer is described by a time-dependent complex number. As explained for complex numbers, the imaginary part and the real part can be separated. The time function is the projection of the pointer onto the real axis. The rotating pointer is described by a time-dependent complex number. As explained for complex numbers, the imaginary part and the real part can be determined separately. The real part and the imaginary part of the complex voltage can be calculated using trigonometric functions.

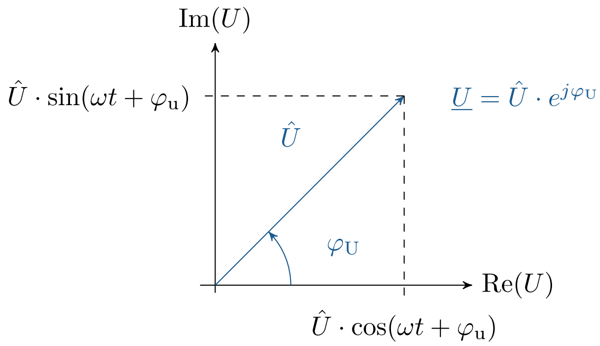


Figure 3.4: **Pointer display of a complex alternating voltage.** Alternating voltage indicator with division into real part and imaginary part.

Key point: Complex rotating pointers for voltage and current

The alternating variables voltage and current are represented as complex rotating pointers. The rotating pointers describe **instantaneous values** $u(t)$, which, with constant **angular velocity** ω change.

Periodic quantities in electrical components are described by fixed voltage indicators and current indicators. The behaviour of the various components differs in each case. Figure 3.5 shows a resistor, a capacitor as capacitance and a coil as inductance with their associated voltage and current vectors. The exact behaviour of the three components will be examined in more detail in the following chapter.

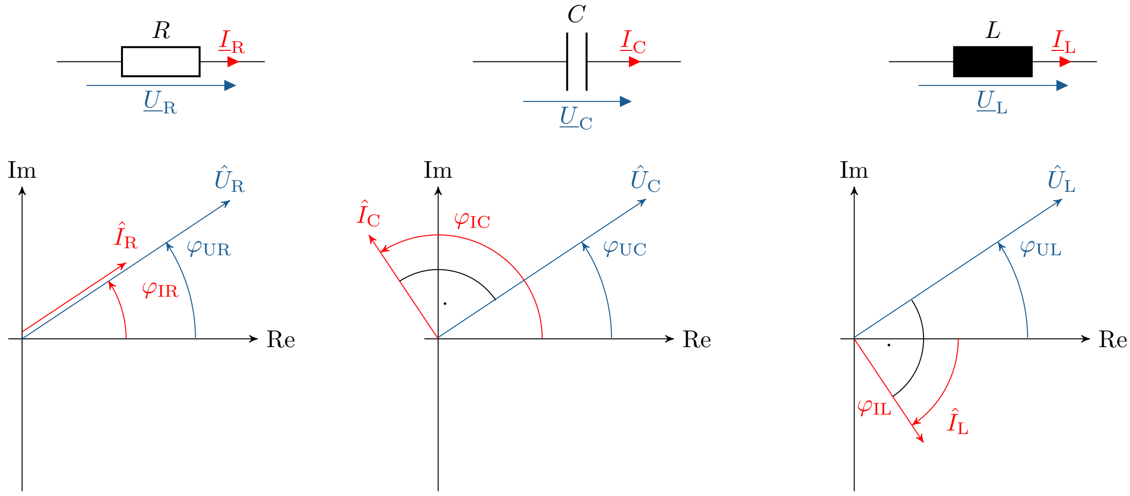


Figure 3.5: **Various components and the alternating voltage indicators.** The components resistor, capacitor and coil with associated voltage and current indicators.

4 Complex alternating current calculation

Direct current networks use constant voltage sources and current sources. In contrast, complex alternating current calculations use sources with sinusoidal alternating quantities. The electrical analysis of components in an alternating voltage network is complex. The complex voltages $\underline{U}$ and currents $\underline{I}$ generate time-dependent voltage values u(t) and current values i(t). Equation 4.1 describes the specified relationship between complex and time-dependent current and voltage values for an ohmic resistor.

Complex and time-dependent current and voltage specifications in AC calculation:

$$\underline{U} = \underline{Z}_{\mathrm{R}} \cdot \underline{I} \quad \rightarrow \quad u(t) = R \cdot i(t) \tag{4.1}$$

Apart from the complexity of alternating current calculations, the rules for calculating electrical networks still apply, such as mesh and node rules, the behaviour of series and parallel circuits, node potential analysis, and mesh current analysis. In all these examples, care must be taken to calculate with complex alternating quantities. For the explanation of electrical networks with complex alternating quantities, the following topics are examined in more detail below: For the calculation of electrical networks with alternating quantities, the following topics are examined in more detail below:

Learning objectives: Alternating current calculation

The students

- know the complex impedance and complex admittance.
- understand the behaviour of a resistor, a capacitor and an inductor at an alternating voltage.
- know alternating current sources and the associated conversion rules.

4.1 Impedance and admittance in the complex plane

A phase shift between the voltage and the current results in the generation of a reactive component of the impedance. However, the reactive component cannot be represented on the real axis, which is why the real axis is extended by an imaginary axis, which together form the complex plane. Figure 4.1 shows the phasor of the complex impedance in the complex plane with imaginary and real components.

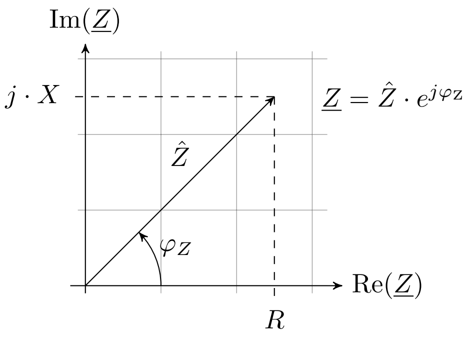


Figure 4.1: **Complex impedance pointer.** The impedance in the complex plane with the division into real part R and imaginary part X.

According to equation 4.2, the complex impedance $\underline{Z}$ also consists of a real part and an imaginary part. The real part consists of the ohmic component, which represents the effective resistance of an impedance. The effective resistance is also referred to as resistance R. The second component of impedance is the imaginary part, which is also referred to as reactance X. The reactance of an electrical network, for example, consists of the capacitive and inductive effects and is assigned the imaginary unit j. As in direct current technology, the unit of complex impedance is the ohm Ω.

$$Impedance = Resistance + j \cdot Reactance \quad \rightarrow \quad \underline{Z} = R + j \cdot X \tag{4.2}$$

$$[\underline{Z}] = 1\ Ohm = 1\ \Omega$$

As in direct current technology, there is a conductance value for electrical resistance in complex alternating current technology. This reciprocal value of the complex impedance is referred to as admittance $\underline{Y}$ (see equation 4.3). The unit of the complex conductance remains Siemens S, as in direct current technology.

$$R = \frac{1}{G} \quad \rightarrow \quad \underline{Z} = \frac{1}{\underline{Y}} \tag{4.3}$$

Admittance can also be differentiated into a real conductance and a reactive conductance. The real conductance G of the admittance is referred to as conductance, and the reactive conductance B of the admittance is referred to as susceptance (see Equation 4.4).

$$Admittance = Conductance + j \cdot Susceptibility \quad \rightarrow \quad \underline{Y} = G + j \cdot B \tag{4.4}$$

$$[\underline{Y}] = 1\ Siemens = 1\ S$$

Key point: Complex impedance and complex admittance

The direct current resistance becomes the complex impedance $\underline{Z}$ for an alternating voltage. The reciprocal of the resistance, the conductance, becomes the complex admittance $\underline{Y}$ for an alternating voltage.

4.2 Complex resistance

The complex resistance of an ohmic consumer consists of the active component R and the reactive component of the ohmic resistance. The ohmic resistance R embodies the property of conductive materials to convert electrical power into thermal power. The reactive component of the resistance does not convert the power of alternating currents into heat. The reactive component of the electrical resistance is denoted as X_R. However, since the ideal electrical resistance is purely resistive and therefore purely real, the symbol R is often used. Equation 4.5 explains these relationships.

$$\underline{Z}_\mathrm{R} = R + j \cdot X_\mathrm{R} \quad \rightarrow \quad X_\mathrm{R-ideal} = j \cdot 0\ \Omega \quad \rightarrow \quad \underline{Z}_\mathrm{R} = R \tag{4.5}$$

The time-dependent values for voltage and current result in the relationships of ideal ohmic resistance according to equation 4.6.

$$\underline{U} = \underline{Z}_\mathrm{R} \cdot \underline{I} \quad \rightarrow \quad u(t) = R \cdot i(t) \tag{4.6}$$

An ideal electrical resistance is a pure active resistance and has no reactive resistance components. Real electrical resistance is subject to parasitic effects. The circuit diagrams of an ideal and a real electrical resistance are compared in Figure 4.2. These parasitic effects have an inductive effect, illustrated by the coil, and a capacitive effect, illustrated by the capacitance, on the circuit. In addition to the frequency dependence of the parasitic effects, the voltage and current are also divided in such a way that the entire power no longer drops across the resistor. The impedance of the ideal electrical resistor is not frequency-dependent.

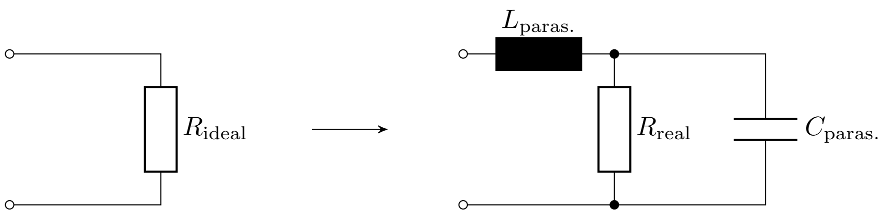


Figure 4.2: **Comparison of ideal and real electrical resistance.** Unlike ideal resistance, real resistance exhibits serial inductive and parallel capacitive effects.

With ideal electrical resistance, there is no phase shift between the voltage and the current. In Figure 4.3, the sine waves for the voltage and current are plotted on an ideal resistor. They lie on top of each other without any phase shift, meaning they are in phase.

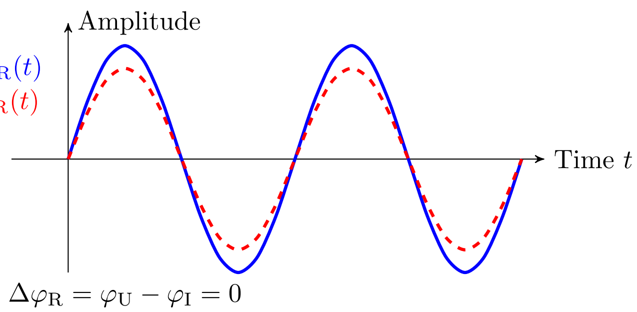


Figure 4.3: **Sine waves on a resistor.** Alternating voltage (blue) and alternating current (red) on an ideal resistor. This results in no phase shift between the alternating voltage and the alternating current at the resistor.

Key point: Sinusoidal oscillation at a resistor

The **electrical resistance** indicates between the alternating voltage and the corresponding alternating current **no phase shift**.

4.3 Capacity

As a component in electrical engineering, the capacitor has the ability to store energy between the two plates in the electric field. The size of a capacitor is specified in terms of its capacitance C. The unit of capacitance is the farad F. The complex impedance of a capacitor can be calculated using equation 4.7. In contrast to the ideal resistor, the ideal capacitor in an alternating current circuit does not act solely as an effective resistance. The real capacitor acts both through an effective component and through an reactive component. The circular frequency ω is frequency-dependent. This results in low impedance values for the capacitor at high frequencies. Conversely, high impedance values result at low frequencies. The reactive component of the capacitor acts as capacitive reactance. The capacitor can also be calculated using complex admittance. The complex admittance is derived from the reciprocal of the complex impedance of the capacitor.

$$\underline{Z}_{\mathrm{C}} = \frac{1}{j\omega C} \quad \rightarrow \quad \underline{Y}_{\mathrm{C}} = j\omega C \tag{4.7}$$

The ideal capacitor acts purely in the reactive component, so according to equation 4.8 from the complex notation Z_C, the reactive component of the capacitor X_C with time-dependent voltage and current.

$$\underline{U} = \underline{Z}_{\mathrm{C}} \cdot \underline{I} \qquad\qquad u(t) = \underline{X}_{\mathrm{C}} \cdot i(t) \tag{4.8}$$

It is assumed that the alternating voltage at a capacitor has a phase shift of 0°. With this assumption, the phase shift of the current at the capacitor can be determined using equation 4.9. Due to the phase shift of the current, the curve of a cosine function contrasts with the sine function of the voltage. If both the voltage and the current are to be described by the sine function, the phase shift $\Delta\varphi$ of $\pi/2$ is included in the argument.

$$u(t) = \hat{U} \cdot \sin(\omega t) \qquad i(t) = \hat{I} \cdot \cos(\omega t) = \hat{I} \cdot \sin(\omega t + \pi/2) \tag{4.9}$$

With inductors and capacitors, there is a phase shift between the alternating voltage and the alternating current. With capacitors, this means that the current leads the voltage by 90° (degrees). The described phenomenon of phase shift is illustrated in Figure 4.4. The current shown in red at a capacitor reaches its maximum value 2π (radians) earlier than the blue-coloured maximum value of the voltage.

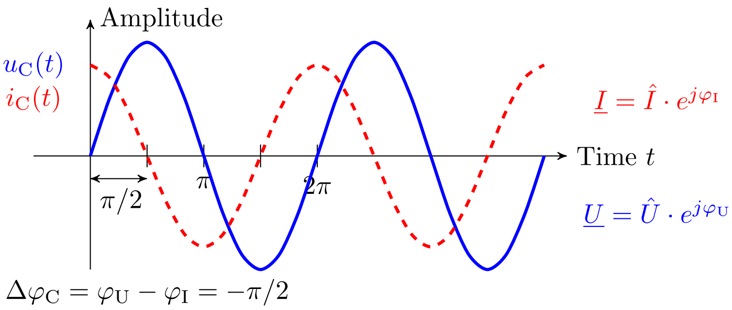


Figure 4.4: **Phase shift between current and voltage at a capacitor.** The current leads the voltage across the capacitor by 90°.

Key point: Sinusoidal oscillation on a capacitor

At the ideal capacity, the alternating current of the alternating voltage leads by 90°.

As a rule, calculations are based on ideal components and their properties. However, real capacitors, like the real resistors described above, exhibit inductive and ohmic parasitic effects. The ideal capacitor is compared with the real capacitor in Figure 4.5. The input and output lines exhibit inductive ($L_{\text{paras.}}$) and ohmic ($R_{\text{paras.2}}$) components. In addition, the capacitor is subject to constant self-discharge. This self-discharge occurs symbolically via the parallel resistance $R_{\text{paras.1}}$.

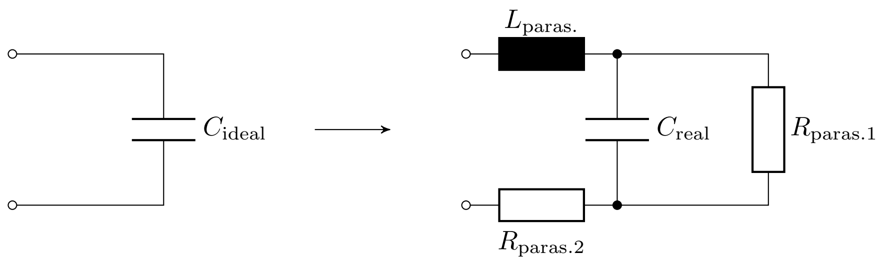


Figure 4.5: **Comparison of an ideal and a real capacitor.** Unlike an ideal resistor, a real capacitor exhibits serial inductive and parallel capacitive effects.

4.4 Inductance

The coil is also an energy storage device, just like the capacitor. However, in the case of the coil, the energy is not stored in an electric field, but in a magnetic field according to the laws of induction. The size of a coil is given by its inductance L. The unit of inductance is Henry H. The complex impedance of a coil is given by the imaginary unit j, the angular frequency ω and the inductance L (see Equation 4.10). The complex admittance is calculated by the reciprocal of the coil impedance. This leads to relationships that develop contrary to the impedance calculations for capacitances. At high frequencies, the impedances of coils also become large in accordance with their inductance and correspondingly small at low frequencies.

$$\underline{Z}_{\text{L}} = j\omega L \quad \rightarrow \quad \underline{Y}_{\text{L}} = \frac{1}{j\omega L} \tag{4.10}$$

As with the capacitor, the ideal coil acts purely as a reactive component of the impedance. In equation

4.11, the complex notation of the coil impedance Z_L is used on the left-hand side. On the right-hand side, the time function is represented solely by the notation of the reactive component of the coil X_L.

$$\underline{U} = \underline{Z}_\mathrm{L} \cdot \underline{I} \qquad\qquad u(t) = \underline{X}_\mathrm{L} \cdot i(t) \tag{4.11}$$

In principle, the time functions of inductances for voltage and current behave similarly to the time functions of capacitances. Only the phase shift of the current in coils is opposite to the phase shift of capacitors. The phase shift $\Delta\varphi$ between the voltage and the current at an inductance is $\pi/2$.

$$u(t) = \hat{U} \cdot \cos(\omega t) = \hat{U} \cdot \sin(\omega t + \pi/2) \qquad i(t) = \hat{I} \cdot \sin(\omega t) \tag{4.12}$$

Equivalent to the phase shift at a capacitor, there is also a phase shift at the inductance. However, this phase shift of the current at the inductance is opposite to the phase shift of the capacitance. In the specific case shown in Figure 4.6, it can be seen that the current (red) at an inductance lags behind the voltage (blue).

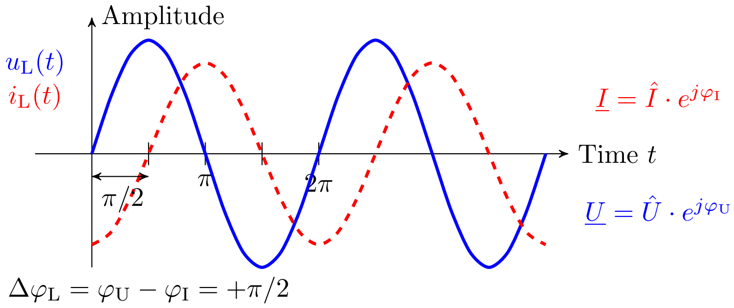


Figure 4.6: **Phase shift between current and voltage at a coil.** The current lags the voltage at the coil by 90°.

Key point: Sinusoidal oscillation on a coil

With ideal inductance, the alternating current lags the alternating voltage by 90°.

Like ohmic resistance and capacitors, real coils exhibit parasitic effects. The coil wire has an ohmic component ($R_\mathrm{paras.}$) that affects the complex impedance of the coil. In addition, the winding of the coil generates capacitive effects ($C_\mathrm{paras.}$). Figure 4.7 shows the equivalent circuit diagram of an ideal coil and a real coil.

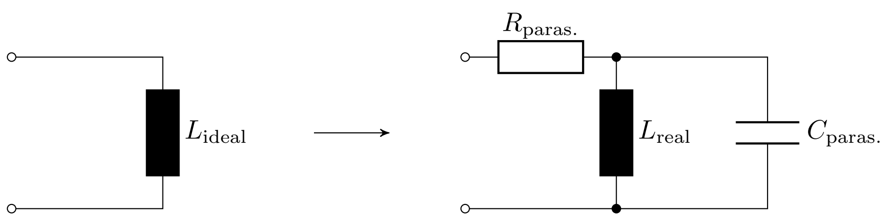


Figure 4.7: **Comparison of an ideal and a real coil.** The real coil exhibits serial ohmic and parallel capacitive parasitic effects.

4.5 Sources of alternating quantities

As with sources of direct current and direct voltage, there are alternating current sources with comparable characteristics. In an ideal alternating voltage source, the voltage is fixed. The alternating voltage does not change depending on the current flowing through it. The ideal alternating current source provides a fixed current . This is not influenced by the connected load. The circuit symbols for an ideal alternating voltage source and an ideal alternating current source are shown in Figure 4.8.

Figure 4.8: **Sources of alternating quantities.** Ideal AC voltage source and ideal AC current source.

The two alternating current sources presented here are ideal sources. Real alternating current sources, like sources of direct voltage and direct current, have internal resistance. This internal resistance manifests itself in sources of alternating quantities as complex impedances. The real alternating voltage source therefore has a complex internal impedance $\underline{Z}_\mathrm{i}$, which is connected in series with an ideal alternating voltage source. In real alternating voltage sources, the output voltage $\underline{U}$ depends on the connected load. The comparison between an ideal and a real alternating voltage source is shown in Figure 4.9.

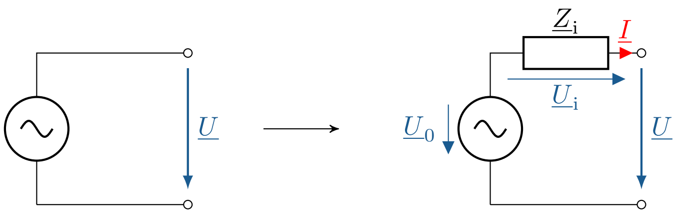


Figure 4.9: **Comparison of an ideal and a real alternating voltage source.** The real alternating voltage source has a series impedance $\underline{Z}_\mathrm{i}$.

The output voltage of a real AC voltage source can be determined using equation 4.13. Here, the AC voltage of the AC voltage source is not directly applied to the terminals. It is reduced by the voltage across the internal impedance Z_i.

$$\underline{U} = \underline{U}_0 - \underline{Z}_\mathrm{i} \cdot \underline{I} \tag{4.13}$$

Real AC voltage sources exhibit certain characteristics when unloaded, at no load and in the event of a short circuit. When idle, there is an infinite impedance between the terminals. No closed circuit is formed and no current flows (equation 4.14). The entire voltage of the AC source is applied to the open terminals, see equation 4.15.

$$\underline{I} = 0 \qquad \rightarrow \qquad \underline{U}_\mathrm{i} = \underline{I} \cdot \underline{Z}_\mathrm{i} = 0 \tag{4.14}$$

$$\underline{U}_0 - \underline{U}_\mathrm{i} - \underline{U} = 0 \qquad \rightarrow \qquad \underline{U} = \underline{U}_0 \tag{4.15}$$

In the event of a short circuit, the impedance between the terminals approaches zero and there is no voltage drop. This results in the maximum possible short-circuit current, which is limited only by the internal impedance (equation 4.16). The voltage of the AC voltage source thus acts in its entirety across the internal impedance (Equation 4.17). The short-circuit current can be determined according to Equation 4.18 from the ratio of the voltage across the internal impedance $\underline{U}_\mathrm{i}$ and the internal impedance $\underline{Z}_\mathrm{i}$.

$$\underline{U} = 0 \qquad \rightarrow \qquad \underline{I} = \underline{I}_\mathrm{k} \tag{4.16}$$

$$\underline{U}_0 - \underline{U}_\mathrm{i} - \underline{U} = 0 \qquad \rightarrow \qquad \underline{U}_\mathrm{i} = \underline{U}_0 \tag{4.17}$$

$$\underline{I}_\mathrm{k} = \frac{\underline{U}_\mathrm{i}}{\underline{Z}_\mathrm{i}} \tag{4.18}$$

Example 4.1: Real Voltage source

A real voltage source (230 V, 50 Hz) has an internal impedance $\underline{Z}_\mathrm{i}$ as shown in the figure below, which consists of a coil $L = 20\ mH$ and an ohmic resistance $R = 10\ m\Omega$.

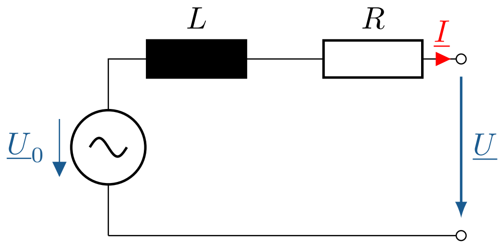


Figure 4.10: **Example.** Alternating voltage source with an internal impedance consisting of a coil and a resistor.

Determine the **open-circuit voltage** and the **short-circuit current** AC voltage source.

$$\begin{aligned}
\underline{U}_0 &= 230\ V \cdot e^{\mathrm{j}0^o} \\
\underline{Z}_\mathrm{i} &= R + jX_L \\
\underline{Z}_\mathrm{i} &= 0{,}01\ \Omega + j \cdot 2\pi \cdot 50\ Hz \cdot 20\ mH \\
\underline{Z}_\mathrm{i} &= 0{,}01\ \Omega + j6{,}28\ \Omega
\end{aligned}$$

Open-circuit voltage:

$$\begin{aligned}
\underline{U} &= \underline{U}_0 \\
\underline{U} &= 230\ V \cdot e^{\mathrm{j}0^o}
\end{aligned}$$

Short-circuit current:

$$\begin{aligned}
\underline{I}_\mathrm{k} &= \frac{\underline{U}_0}{\underline{Z}_\mathrm{i}} \\
\underline{I}_\mathrm{k} &= \frac{230\ V \cdot e^{\mathrm{j}0^o}}{0{,}01\ \Omega + j6{,}28\ \Omega} \\
\underline{I}_\mathrm{k} &= (0{,}058 - j36{,}6)\ A = 36{,}6\ A \cdot e^{-\mathrm{j}89{,}9^o}
\end{aligned}$$

Like the AC voltage source, the real AC source in Figure 4.11 has an internal impedance $\underline{Z}_{\mathrm{i}}$, whereby this internal impedance is not connected in series as in the AC voltage source, but rather in parallel to an ideal AC source. The resulting output current $\underline{I}$ of the source depends on the connected load or the output voltage $\underline{U}$.

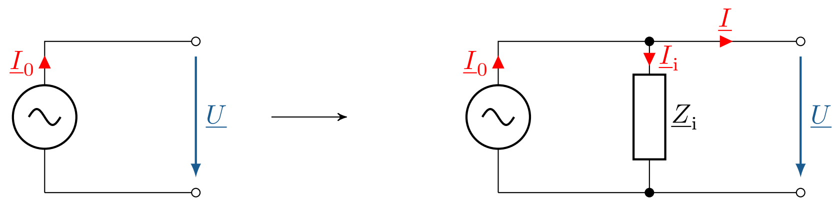


Figure 4.11: **Sources of alternating quantities.** Ideal AC source and ideal AC source.

The alternating current of the ideal alternating current source is reduced by a portion due to the internal impedance. The output current of the real alternating current source can be determined using equation 4.19. Here, starting from the source current $\underline{I}_0$, the output voltage-dependent ($\underline{U}$) alternating current component $\underline{I}_{\mathrm{i}}$ is subtracted by the internal impedance $\underline{Z}_{\mathrm{i}}$.

$$\underline{I} = \underline{I}_0 - \underline{I}_{\mathrm{i}} = \underline{I}_0 - \frac{\underline{U}}{\underline{Z}_{\mathrm{i}}} \tag{4.19}$$

AC conversion

Alternating current and alternating voltage sources can be converted into each other if both alternating sources have the same short-circuit current and the same open-circuit voltage.

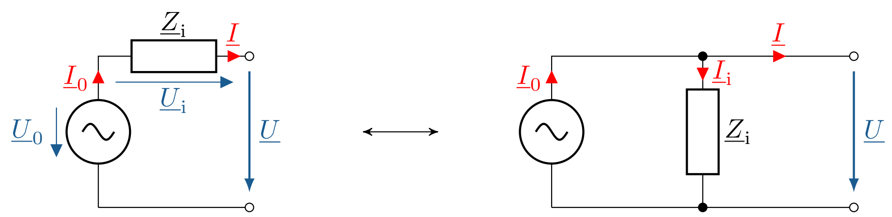


Figure 4.12: **Comparison of a real alternating voltage source and a real alternating current source.** The real AC voltage source and the real AC current source are convertible into each other.

Both AC sources are equivalent if, according to Equation 4.20 and Equation 4.21, Ohm's law applies with identical internal impedances for the AC voltage source and the AC current source.

$$\underline{U}_0 = \underline{Z}_0 \cdot \underline{I}_0 \tag{4.20}$$

$$\underline{Z}_{\mathrm{i}} = \underline{Z}_0 \tag{4.21}$$

Here, the short-circuit current $\underline{I}_{\mathrm{K}}$ should be equal to the source current $\underline{I}_0$ of the alternating current source, which is identical to the output current $\underline{I}$. The output alternating voltage $\underline{U}$ of the alternating sources corresponds to the open-circuit voltage $\underline{U}_{\mathrm{L}}$.

$$\underline{I} = \underline{I}_{\mathrm{k}} = \underline{I}_0 \qquad \text{und} \qquad \underline{U} = \underline{U}_{\mathrm{L}} = \underline{U}_0 \tag{4.22}$$

Example 4.2: AC conversion

Given is a real AC voltage source with $\underline{U}_0 = 20\ kV \cdot \mathrm{e}^{\mathrm{j}30^o}$ and $\underline{Z}_0 = (0.01 + j \cdot 2)\ \Omega$. Convert the alternating voltage source into an equivalent alternating current source and calculate the short-circuit current.

Short-circuit current:

$$\underline{I}_0 = \frac{\underline{U}_0}{\underline{Z}_0} = \frac{20\ kV \cdot \mathrm{e}^{\mathrm{j}30^o}}{(0{,}01 + j \cdot 2)\ \Omega} = (5{,}04 + j \cdot 8{,}64)\ kA$$

4.6 Complex voltage divider and complex current divider

As with the consideration of direct current networks, the basic laws for analysing alternating current networks apply. The node rule and the mesh rule are to be applied equivalently here. The example node and the example mesh already presented are shown once again in Figure 4.13 with complex alternating quantities. This results in the formulation of the following equation 4.23 as an example of the complex node rule:

$$\sum_{k=1}^{N} \underline{I}_{\mathrm{k}} = 0 = \underline{I}_1 + \underline{I}_2 - \underline{I}_3 - \underline{I}_4 - \underline{I}_5 \tag{4.23}$$

As well as equation 4.24 as an example of the complex mesh rule:

$$\sum_{\mathrm{k}=1}^{N} \underline{U}_{\mathrm{k}} = \underline{U}_{\mathrm{Z2}} - \underline{U}_{\mathrm{Z1}} = 0 \tag{4.24}$$

In contrast to direct current analysis, where only real values are used in calculations, here it is only necessary to ensure that complex alternating quantities are used.

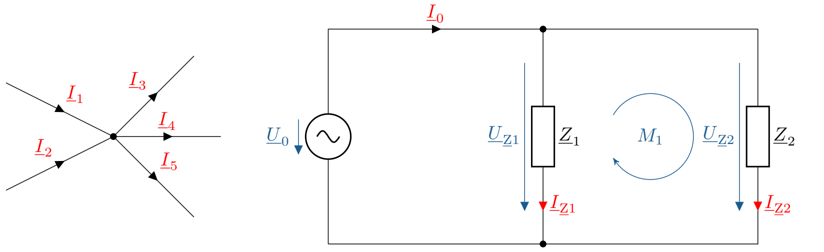


Figure 4.13: **Knot rule and mesh rule with complex alternating variables.** The node rule and the mesh rule behave identically with complex alternating quantities as with direct current networks.

The complex voltage divider is used to reduce a total voltage, e.g. to adjust a signal level to the input voltage range of a measuring circuit. Figure 4.14 shows a complex voltage divider consisting of two complex impedances. The complex total voltage $\underline{U}_0$ is divided into the partial voltages $\underline{U}_1$ and $\underline{U}_2$ via the two impedances $\underline{Z}_1$ and $\underline{Z}_2$.

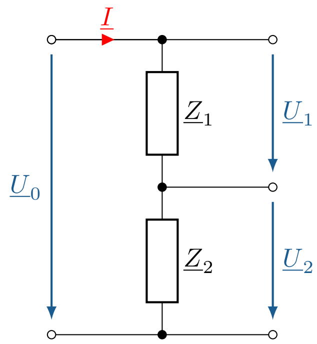


Figure 4.14: **A complex voltage divider consisting of two impedances in series.** The complex total voltage $\underline{U}_0$ is divided into the partial voltages $\underline{U}_1$ and $\underline{U}_2$

The total voltage is divided in proportion to the complex impedances (see Equation 4.25). Based on the complex current $\underline{I}$ flowing through the entire series connection, the complex partial voltages $\underline{U}_1$ and $\underline{U}_2$ and their complex impedances $\underline{Z}_1$ and $\underline{Z}_2$ can be determined in relation to the complex total voltage $\underline{U}_0$ and the total complex impedance $\underline{Z}_1 + \underline{Z}_2$ can be determined.

$$\underline{I} = \frac{\underline{U}_0}{\underline{Z}_1 + \underline{Z}_2} = \frac{\underline{U}_1}{\underline{Z}_1} = \frac{\underline{U}_2}{\underline{Z}_2} \tag{4.25}$$

By rearranging equation 4.25, the complex partial ratio $\underline{T}$ for equation 4.26 can be established:

$$\underline{T} = \frac{\underline{U}_2}{\underline{U}_0} = \frac{\underline{Z}_2}{\underline{Z}_1 + \underline{Z}_2} \tag{4.26}$$

Here, the ratio of the complex partial voltage $\underline{U}_2$ and the complex total voltage $\underline{U}_0$ corresponds to the ratio of the complex impedance $\underline{Z}_2$ and the total complex impedance of the series connection $\underline{Z}_1+\underline{Z}_2$. The complex partial ratio can be established in the same way for the complex partial voltage $\underline{U}_1$.

The complex voltage divider is used, for example, in ohmic-capacitive dividers in high-voltage technology. Figure 4.15 shows four different dividers that are used, for example, in high-voltage technology.

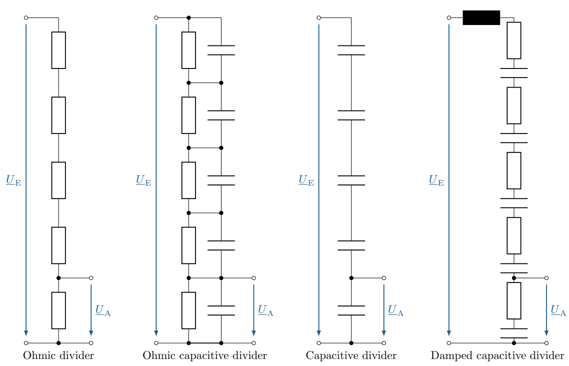


Figure 4.15: **Comparison of different dividers in high-voltage technology.** Ohmic divider for use with direct voltage and capacitive divider and damped capacitive divider for use with alternating voltage. The ohmic-capacitive divider can be used for both types of voltage.

In addition to the ohmic-capacitive divider, the ohmic divider, the capacitive divider and the damped capacitive divider are also presented. The ohmic divider consists purely of ohmic resistors, which are connected in series, and the capacitive divider consists purely of capacitors. The damped capacitive divider consists of several modules with an upstream coil. The modules are composed of series connections of an resistive resistor and a capacitor.

A simple example of such an ohmic-capacitive divider is shown in Figure 4.16. The ohmic-capacitive divider consists of at least two parallel circuits connected in series, each comprising an ohmic resistor and a capacitor, known as an RC element . In Figure 4.16, the first RC element consists of the resistor R_1 and the capacitor C_1. The second RC element consists of the resistor R_2 and the capacitor C_2. The complex total voltage $\underline{U}_0$ is applied across both RC elements. The complex partial voltage $\underline{U}_2$ is applied across the second RC element. The partial ratio therefore has ohmic and capacitive components. The capacitance of the capacitor results in a frequency-dependent partial ratio.

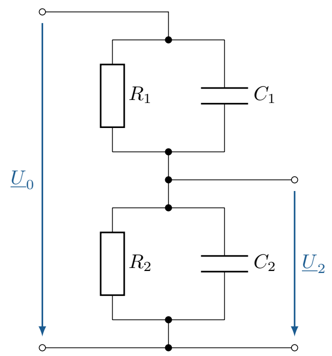


Figure 4.16: **An ohmic-capacitive voltage divider consisting of two RC elements.** The RC elements each consist of a parallel connection of a resistor and a capacitor.

A special condition exists when the complex voltage divider is balanced. This condition describes the identical ratio of the components to each other. This condition is described by equation 4.27. Here the product of the resistor R_1 and the capacitor C_1 must be identical to the product of the resistor R_2 and the capacitor C_2. In practice, this balancing condition can be achieved using variable components such as trimmer capacitors.

$$R_1 \cdot C_1 = R_2 \cdot C_2 \tag{4.27}$$

Another example of an ohmic-capacitive divider is the probe of an oscilloscope. Such a divider is shown in Figure 4.17. In principle, the image does not differ significantly from the ohmic-capacitive divider. However, here the first and second RC elements are spatially separated from each other and enclosed in a housing. The first RC element represents the probe and the second RC element represents the circuitry in the oscilloscope. With the ohmic division ratio:

$$\frac{\underline{U}_\mathrm{A}}{\underline{U}_\mathrm{E}} = \frac{R_2}{R_1 + R_2} \tag{4.28}$$

and the capacitive ratio:

$$\frac{\underline{U}_\mathrm{A}}{\underline{U}_\mathrm{E}} = \frac{C_2}{C_1 + C_2} \tag{4.29}$$

To calibrate the probe, the ohmic and capacitive division ratios can be equated, as the division ratios must be identical:

$$\frac{R_2}{R_1 + R_2} = \frac{C_2}{C_1 + C_2} \tag{4.30}$$

The adjustment condition according to equation 4.27 still applies. The divider is adjusted via the adjustable capacitance C_1.

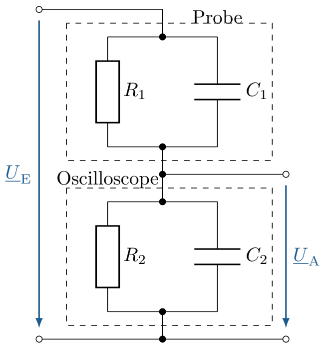


Figure 4.17: **An ohmic-capacitive voltage divider as the probe head of an oscilloscope.** The complex voltage divider is balanced via capacitor C_1. The voltage divider is balanced via capacitor C_1.

The complex current divider, as shown in Figure 4.18, divides the complex total current $\underline{I}$ into the partial currents $\underline{I}_1$ and $\underline{I}_2$. The two partial currents $\underline{I}_1$ and $\underline{I}_2$ flow through the admittances $\underline{Y}_1$ and $\underline{Y}_2$.

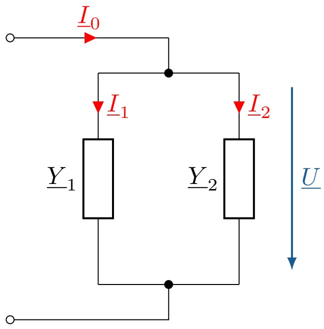


Figure 4.18: **A complex current divider consisting of two admittances.** The complex current $\underline{I}$ is divided into the currents $\underline{I}_1$ and $\underline{I}_2$.

The complex voltage $\underline{U}$ can be calculated according to Equation 4.31 using the ratio of the total current $\underline{I}$ and the total admittance $\underline{Y}_1 + \underline{Y}_2$ or using the ratios of the individual current branches:

$$\underline{U} = \frac{\underline{I}_0}{\underline{Y}_1 + \underline{Y}_2} = \frac{\underline{I}_1}{\underline{Y}_1} = \frac{\underline{I}_2}{\underline{Y}_2} \tag{4.31}$$

The division ratio can be defined for the two current branches according to equation 4.32:

$$\underline{T}_{\mathrm{i1}} = \frac{\underline{I}_1}{\underline{I}_0} = \frac{\underline{Y}_1}{\underline{Y}_1 + \underline{Y}_2} \tag{4.32}$$

$$\underline{T}_{\mathrm{i2}} = \frac{\underline{I}_2}{\underline{I}_0} = \frac{\underline{Y}_2}{\underline{Y}_1 + \underline{Y}_2} \tag{4.33}$$

5 RMS value

Due to the periodic voltage curves, each time step has a different voltage value. In this case, only the peak value can be specified directly, but this cannot always be used for calculations. To ensure useful comparability of voltage values, the effective value is used. The RMS value describes, for example, the identical power that would be generated at a resistor if it were connected to a direct current source. For a further explanation of the RMS value and the necessary fundamentals, this chapter discusses the following topics:

Learning objectives: RMS value

The students

- understand the function of the effective value.
- are familiar with the effects of amplitude and waveform on the effective value.
- can calculate the effective value of an alternating voltage or alternating current.

5.1 Fundamentals: Mean square and addition theorem

In addition to the arithmetic mean, which is used in alternating current calculations to determine the mean value, the quadratic mean is used to determine the RMS value. The quadratic mean (RMS – root mean square) is used to enable the comparability of alternating voltage values. First, a function f(t) is squared. Then, the mean value of the squared function is calculated. The final step is to take the square root of the calculated mean value. The mathematical relationship of the RMS is shown in equation 5.1.

$$RMS = \sqrt{\frac{1}{T}\int_0^T f(t)^2 dt} \tag{5.1}$$

When calculating the quadratic mean, the sine function is also squared. To assist with this, an addition theorem is presented to simplify the calculation. Addition theorems refer to the relationships in trigonometry used to transform a function. The transformation of the quadratic sine function using the addition theorem is shown in equation 5.2. Here, the quadratic sine function becomes a function that depends only on a simple cosine function.

$$\sin^2(x) = \frac{1}{2}(1 - \cos(2x)) \tag{5.2}$$

5.2 Amplitude

The amplitude of a function describes the maximum deflection during one period. For an alternating voltage, the amplitude value indicates the highest instantaneous voltage value. In Figure 5.1, voltage signals with a frequency of $50\,Hz$ and thus with an identical period duration of $T = 20\,ms$ are shown. They differ in their amplitude values. The peak values of the amplitudes of the voltages shown are $\hat{U}_1 = 1\,V$, $\hat{U}_2 = 2\,V$ and $\hat{U}_3 = 4\,V$. The amplitude is a kind of factor for calculating the RMS value. Thus, the height of the RMS value depends directly on the amplitude value.

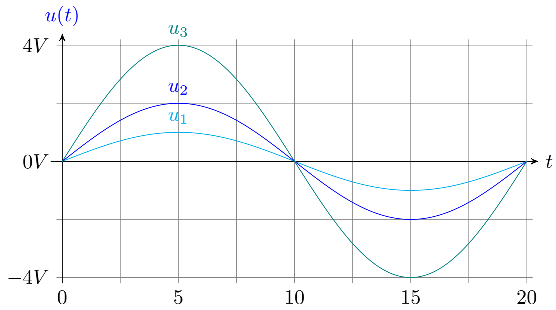


Figure 5.1: **Sinusoidal voltages with different amplitude values.** Weighting of amplitude values $\hat{U}_1 = 1\,V$, $\hat{U}_2 = 2\,V$ and $\hat{U}_3 = 4\,V$.

5.3 Waveform

Neben der Amplitude eines Signals ist die Kurvenform des Signals ausschlaggebend für den Effektivwert. Bei der Berechung des Effektivwerts wird der Scheitelfaktor C (engl. für crest factor) bestimmt, welcher mit dem Ampltudenwert verrechnet wird. In der Gleichung 5.3 wird die Beziehung zwischen dem Amplitudenwert und den Effektivwert über den Scheitelfaktor angegeben.

$$\hat{U} = C \cdot U_{\text{RMS}} \tag{5.3}$$

This crest factor is $C_S = \sqrt{2}$ for sine waves, $C_D = \sqrt{3}$ for triangular waves, and $C_R = 1$ for square waves. Thus, the RMS value of a sine wave voltage is approximately 70.7 per cent of the peak value of the amplitude. In addition to the crest factor as the relationship between the peak value and the effective value, the form factor F is also used. The form factor is defined as the quotient of the RMS value and the rectified value.

Table 5.1: **Waveform shapes and crest factors** The crest factors for sinusoidal, triangular and rectangular signals.

Waveform:	Sinus	Triangle	Rectangle
Crest factor (C):	$\sqrt{2}$	$\sqrt{3}$	1

Key point: Influences on the RMS value

The RMS value is linearly proportional to the peak value. The conversion factor from the RMS value to the peak value is called the crest factor and depends on the waveform of the signal.

5.4 RMS value

The RMS value, for example of a sinusoidal voltage, should be used for the power consumption as an instantaneous value that causes the same power consumption that would correspond to the power value of a DC voltage under comparable conditions.

The RMS value is also applicable to Ohm's law and Kirchhoff's laws and is therefore suitable for all fundamental laws. The calculation of the RMS value will be explained using an alternating current signal $i(t)$. If the definition of the current signal is inserted into equation 5.4, the result is:

$$f(t) = i(t) = \hat{I} \cdot \sin(\omega t) \quad \rightarrow \quad I_{\text{Eff}} = \sqrt{\frac{1}{T} \int_0^T \hat{I}^2 \cdot \sin^2(\omega t) dt} \tag{5.4}$$

The square root of the peak value is not time-dependent and can be moved outside the integral sign. In addition, the factor from the addition theorem can be removed from the integral, resulting in the following equation 5.5:

$$I_{\text{RMS}} = \sqrt{\frac{\hat{I}^2}{2T} \int_0^T 1 - \cos(2\omega t) dt} \tag{5.5}$$

According to the difference rule, differences in an integrand can be determined separately. In this way in the following equation 5.6, the integrals for the two operands of the difference are determined separately.

$$\text{with} \quad \int_0^T 1 dt = T \quad \text{and} \quad \int_0^T \cos(2\omega t) dt = 0 \tag{5.6}$$

Solving the integral yields:

$$I_{\text{RMS}} = \sqrt{\frac{\hat{I}^2}{2T} \cdot (T - 0)} \tag{5.7}$$

Here, the period duration T can be reduced from the numerator and denominator, resulting in the following relationship:

$$I_{\text{RMS}} = \sqrt{\frac{\hat{I}^2}{2}} \tag{5.8}$$

Now the root is taken separately from the numerator and denominator. The same relationship can also be defined for the voltage:

$$I_{\text{RMS}} = \frac{\hat{I}}{\sqrt{2}} \quad \text{and} \quad U_{\text{RMS}} = \frac{\hat{U}}{\sqrt{2}} \tag{5.9}$$

Key point: RMS value of a sinusoidal oscillation

For purely sinusoidal voltages and currents, the crest factor $\sqrt{2}$ applies:

$$I_{\text{RMS}} = \frac{\hat{I}}{\sqrt{2}} \quad \text{and} \quad U_{\text{RMS}} = \frac{\hat{U}}{\sqrt{2}}$$

Figure 5.2 shows a sinusoidal voltage with an amplitude value of $4\,V$. The RMS value associated with the sinusoidal voltage is plotted at $4\,V/\sqrt{2}$. The situation is similar with voltages in the home. For example, a voltage measurement in the household would show an RMS value of 230 V. However, the amplitude value of the actual alternating voltage is $230\,V \cdot \sqrt{2}$, i.e. 325 V.

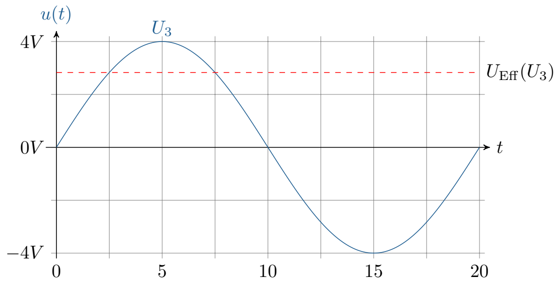


Figure 5.2: **Waveform of a sinusoidal voltage.** Sine voltage with an amplitude value of $4\,V$ and corresponding RMS value of $4\,V/\sqrt{2}$.

Example 5.1: RMS value

If the voltage is measured at a domestic socket, for example, the RMS value of 230V is specified.

The following tasks should be completed to gain a better understanding of the RMS value:

a) Calculation of the amplitude value of the voltage at a domestic power socket.

b) Calculation of the RMS value assuming that it is a rectangular signal.

a) The amplitude value at a socket with an RMS value of 230V is:

$$\hat{U} = I_{\text{RMS}} \cdot \sqrt{2} = 230V \cdot \sqrt{2}$$
$$\hat{U} = 325{,}269V$$

b) The RMS value of a square wave voltage with a peak value of 325V is:

$$U_{\text{Rectangle}} = \frac{\hat{U}}{\sqrt{3}} = \frac{325V}{\sqrt{3}}$$
$$U_{\text{Rectangle}} = 187{,}639V$$

6 Apparent power, active power and reactive power

In the direct current analysis of electrical networks, only active power was found at the components. In the alternating current analysis of electrical networks, the resulting reactive currents also generate reactive power. The resulting apparent power is divided between active power and reactive power. The following fundamentals and types of power caused by alternating voltage are discussed below:

Learning objectives: Apparent power, active power and reactive power

The students

- know the difference between instantaneous power, active power and reactive power.
- can determine and categorise apparent power.

6.1 Fundamentals: Addition theorem and arithmetic mean

An addition theorem has already been presented for calculating the RMS value. Another addition theorem is presented in equation 6.1. Here, the product of two sine functions with different arguments is related to sums and differences in cosine functions. This eliminates the product and creates a sum of two cosine functions. Integrating the sum can be simplified by dividing the sum into two separate integration processes.

$$\sin(x) \cdot \sin(y) = \frac{1}{2}(\cos(x-y) + cos(x+y)) \tag{6.1}$$

The arithmetic mean of a function is used to calculate a constant value that is no longer time-dependent. In electrical engineering, this value is used in certain cases, similar to the RMS value for sinusoidal current and voltage curves, for better illustration. The arithmetic mean is calculated using the following integral formula:

$$\overline{u(t)} = \frac{1}{T} \int\limits_{t}^{t+T} u(t) dt \tag{6.2}$$

In the formula, T stands for the period length. Here, t is any freely selectable point in time from which the observation should start (usually $t = 0$). Put simply, the integral represents the area between the function curve and the abscissa and the denominator of the fraction represents the period length. As an example, let us consider a so-called sawtooth voltage, which is shown in Figure 6.1.

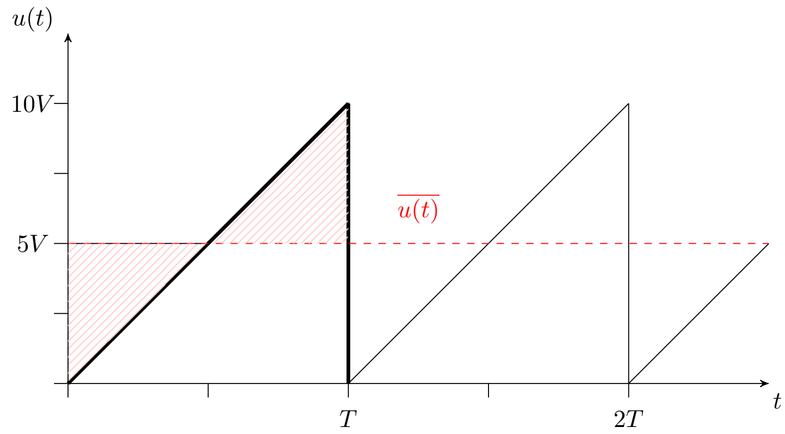


Figure 6.1: **Progression of a sawtooth wave.** The sawtooth voltage is used to explain how to determine the mean value marked in red.

To calculate the arithmetic mean, a function must first be established over a period. In this case, a period can be expressed as follows:

$$\overline{u(t)} = \frac{10\ V}{T} \cdot t \tag{6.3}$$

Inserting this relationship into equation 6.2 and setting the integration limits $t = 0$ to T yields the equation:

$$\overline{u(t)} = \frac{1}{T} \int_0^T \frac{10\ V}{T} \cdot t\ dt \tag{6.4}$$

When the integral is resolved, the arithmetic mean value for the sawtooth voltage is obtained:

$$\overline{u(t)} = \frac{1}{T} \left[\frac{10\ V}{T} \cdot \frac{t^2}{2} \right]_0^T = \frac{1}{2} \cdot 10\ V \tag{6.5}$$

The arithmetic mean is therefore half the maximum voltage. In the case of an ideal sine wave, the arithmetic mean is always zero. The arithmetic mean is relevant when the power of sinusoidal currents and voltages is to be calculated.

6.2 Active power and reactive power

Active power in direct current: In direct current electrical networks, power is determined using the well-known equation $P = U \cdot I$. If the electrical resistance is known, the equation can be rearranged so that, according to Ohm's law, either the voltage or the current is eliminated. Active power in alternating current:

$$P = U \cdot I = I^2 \cdot R = \frac{U^2}{R} \tag{6.6}$$

In a system with alternating quantities, power, like current and voltage, is time-dependent. In equation 6.7, this is expressed by the fact that the symbols are written in lower case and are dependent on time t:

$$P = U \cdot I \quad \Leftrightarrow \quad p(t) = u(t) \cdot i(t) \tag{6.7}$$

Because the magnitude of voltage and current are time-dependent, this function of power is also time-dependent. Due to its time dependency, we also refer to **Augenblicksleistung** for a given point in time t. The unit of measurement for instantaneous electrical power continues to be the watt.

$$u(t) = \hat{U} \cdot \sin(\omega t + \varphi_{\mathrm{U}}) \qquad i(t) = \hat{I} \cdot \sin(\omega t + \varphi_{\mathrm{I}}) \tag{6.8}$$

If the instantaneous values of voltage and current are substituted into equation 6.7 for instantaneous power, the following equation 6.10 gives the instantaneous power. Here, the equation can be divided into a leading factor, a constant component and a time-dependent component. The pre-factor consists of half the product of the amplitude values of voltage and current. The constant part consists of the first cosine summand; here, only the phase angles of voltage and current are decisive. The time-dependent part is represented by the second cosine. Here, in addition to the phase angles, the angular frequency and the time are also important.

$$\text{Generally:} \quad p(t) = u(t) \cdot i(t) \tag{6.9}$$

$$\text{Sinusoidal:} \quad p(t) = \frac{\hat{U} \cdot \hat{I}}{2}(\underbrace{\cos(\varphi_{\mathrm{U}} - \varphi_{\mathrm{I}})}_{\text{constant proportion}} + \underbrace{\cos(2\omega t + \varphi_{\mathrm{U}} + \varphi_{\mathrm{I}})}_{\text{time-dependent portion}}) \tag{6.10}$$

Key point: Instantaneous power

The instantaneous power is the product of the instantaneous values of voltage and current.

$$p(t) = u(t) \cdot i(t)$$

If the entire power of a system is absorbed via an ideal ohmic resistor, this means that the voltage in the current has no phase shift. The constant component of the instantaneous power is positive. The instantaneous power pulsates at twice the frequency compared to the frequency of the voltage and current. In this case, the power always has a positive sign.

Often, the exact temporal course of the instantaneous power is not significant. Rather, in applications only the average power converted, for example heat applications, is of interest. The average power of the time-varying instantaneous power is defined by equation 6.2. For purely sinusoidal voltages and currents, the average power can be determined using equation 6.11. The voltage U and current I values are the RMS value. The angle φ results from the phase shift between the voltage and the current. The entire expression $\cos(\varphi)$ is also referred to as the **Leistungsfaktor**.

$$\overline{p}_{\mathrm{Sin}} = P = U \cdot I \cdot \cos(\varphi) \tag{6.11}$$

Without the phase shift between the voltage and the current at an ideal resistor, the power factor term is negligible. Thus, in alternating current technology, the average power can be reduced to two time-independent variables in the case of sinusoidal voltage and current curves, analogous to direct current technology. The RMS value of the voltage and current are used to determine the average power for purely sinusoidal quantities.

$$\overline{p}_{\mathrm{Ohm}} = \frac{\hat{U}}{\sqrt{2}} \cdot \frac{\hat{I}}{\sqrt{2}} = \frac{\hat{U} \cdot \hat{I}}{2} = U \cdot I \tag{6.12}$$

The situation is similar with the instantaneous power at coils and capacitors. At an inductance, the current lags the voltage by 90° ($+\pi/2$). The voltage and current waveforms across an inductance are shown in Figure 6.2. The instantaneous power across the inductance is shown in green. The constant component of the instantaneous power across an inductance is zero. The curve shows the pulsation of the instantaneous voltage with twice the frequency of the voltage or current. The positive and negative half-waves of the instantaneous power cancel each other out on average. This means that the power consumed is completely re-emitted and thus no power is consumed by the inductance.

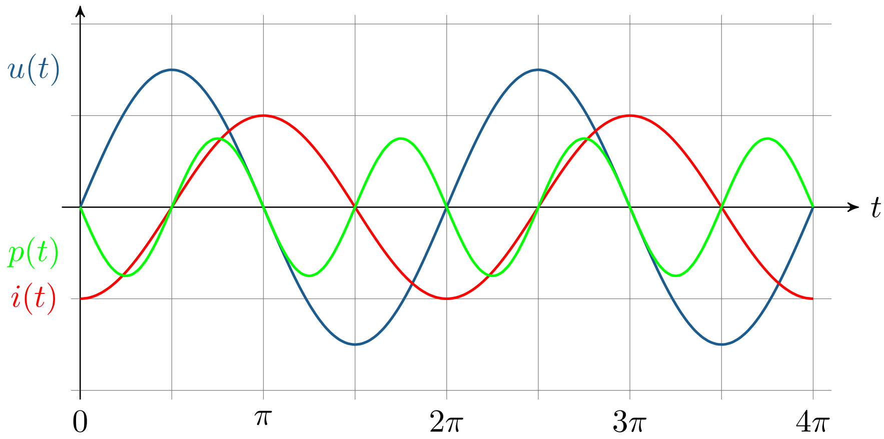


Figure 6.2: **Curve of current, voltage and time-dependent power at an inductive load.** The instantaneous power has twice the frequency of the voltage or current. The positive and negative half-waves cancel each other out, and the power consumed is completely re-emitted.

In principle, the capacitor behaves in the same way as inductance. Here, the current lags the voltage by 90° ($-\pi/2$). The constant component of the instantaneous power is zero and the power pulsates at twice the frequency around the zero line. As with inductance, the power absorbed by a capacitor is completely re-emitted, so that on average no power is absorbed. The behaviour of time-dependent power at a capacitive load is shown in Figure 6.3.

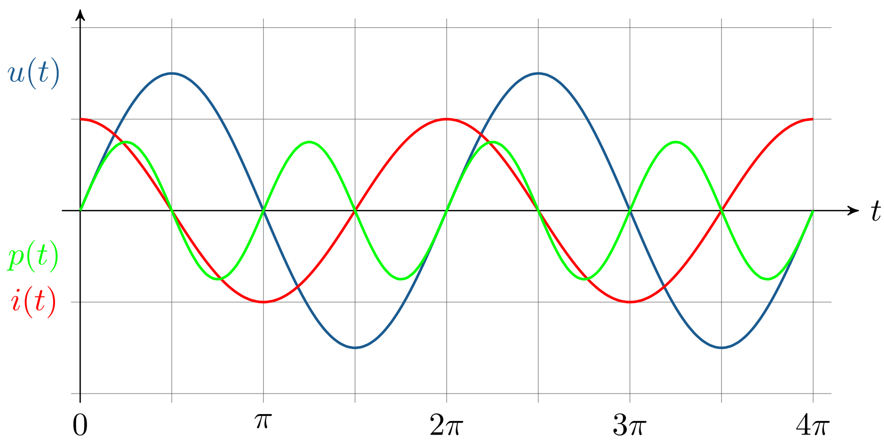


Figure 6.3: **Curve of current, voltage and time-dependent power at a capacitive load.** The instantaneous power has twice the frequency of the voltage or current. The positive and negative half-waves cancel each other out, and the power consumed is completely re-emitted.

If the instantaneous power is greater than zero, power is absorbed. Conversely, power is delivered when the instantaneous power is less than zero. Due to the absence of phase shift between the current and voltage at the ohmic resistor, the average power is always greater than zero. The ohmic resistance therefore always consumes power. The power is referred to here as active power P, which can be used for applications. It is calculated using the equation 6.11 already presented for sinusoidal voltages and currents. The powers described explain active power components and reactive power components. Reactive power describes the portion of power that is necessary for the generation of electric and magnetic fields. If the electrical impedance is purely inductive or capacitive, there is a phase shift of $\pm 90°$. The purely inductive or purely capacitive two-pole absorbs power, which is then completely released again. The resulting mean power is always zero. This power is referred to as reactive power Q. A distinction is made between inductive and capacitive reactive power. For inductive reactive power, there is a positive phase shift for the power, which corresponds exactly to the course of the sine function. The capacitive reactive power is shifted in the opposite direction, so that a sine function can also be defined, but with a negative sign. Equations 6.13 and 6.14 present the equations for calculating inductive and capacitive reactive power. The unit of measurement for inductive and capacitive reactive power is the var (volt-ampere reactive).

$$Q_{\text{Ind}} = U \cdot I \cos(\varphi + 90°) = U \cdot I \cdot \sin(\varphi) \tag{6.13}$$

$$Q_{\text{Kap}} = U \cdot I \cos(\varphi - 90°) = -U \cdot I \cdot \sin(\varphi) \tag{6.14}$$

$$[Q] = 1\ Volt - Ampere - reactive = 1\ \text{var}$$

With ideal capacitive or inductive resistors, no energy is converted into heat. Only the magnetic or electric field changes, so energy is stored or released.

Key point: Active power and reactive power

At the complex level, power is explained by active power and reactive power. Active power describes the real part and reactive power the imaginary part of the complex apparent power. Active power and reactive power are defined as follows:

$$P = U \cdot I \cdot \cos(\varphi) \qquad Q = U \cdot I \cdot \sin(\varphi)$$

6.3 Apparent power

So far, we have looked at the basics of power analysis for ideal ohmic and inductive/capacitive impedances. However, most networks contain both ohmic and inductive or capacitive impedances. In such networks, a distinction is made between apparent power, active power and reactive power. The apparent power represents the total power of the system. To calculate the apparent power, the maximum value is used, i.e. when there is no phase shift. This value is therefore equal to the power of an ideal ohmic resistor:

$$S = U \cdot I = P_{\text{R}} \tag{6.15}$$

If the apparent power is considered in the complex plane, the fixed pointers are used. If the complex voltage is multiplied by the complex current, the phase angles would be added according to the rules of complex arithmetic. However, in order to perform a correct power calculation, there must be a difference between the phase angles. To remedy this problem, the conjugate complex value of the current is used in the calculation:

$$\begin{aligned}
& \underline{S} = \underline{U} \cdot \underline{I}^* \\
\textit{with} \quad & \underline{U} = U \cdot e^{j\varphi_\mathrm{u}} \\
\textit{and} \quad & \underline{I}^* = I \cdot e^{-j\varphi_\mathrm{i}} && (6.16) \\
\textit{follows} \quad & \underline{S} = U \cdot e^{j\varphi_\mathrm{u}} \cdot I \cdot e^{-j\varphi_\mathrm{i}} \\
& \quad = U \cdot I \cdot e^{j(\varphi_\mathrm{u} - \varphi_\mathrm{i})} && (6.17) \\
& \quad = U \cdot I \cdot e^{j(\varphi)}
\end{aligned}$$

The complex value of the voltage and the conjugate complex value of the current can be used to determine the active power and the reactive power separately, or the complex apparent power directly:

$$U \cdot I \cdot e^{j\varphi} = U \cdot I \cdot (\cos\varphi + j\ \sin\varphi) = P + jQ = \underline{S} \tag{6.18}$$

In addition, the power factor can also be used to directly determine a relationship between the active power or reactive power and the apparent power according to Equation 6.19.

$$P = \cos(\varphi) \cdot S \qquad\qquad Q = S \cdot \sin(\varphi) \tag{6.19}$$

Key point: Komplexe Scheinleistung

The complex apparent power is composed of active power and reactive power. It can be determined in various ways:

$$\underline{S} = \underline{U} \cdot \underline{I}^* = U \cdot I \cdot e^{j(\varphi_\mathrm{u} - \varphi_\mathrm{i})} = P + jQ$$

The quantities active power and reactive power are found in the complex number plane. Active power is a purely real value, while reactive power is a purely imaginary value. Apparent power, which is also a complex quantity, can be represented as the sum of active and reactive power (see Figure 6.4). The apparent power S shown here with the associated phase angle φ is located in the I quadrant. This is caused by a positive active power component and a positive reactive power component. The positive active power indicates that some electrical power is being consumed, i.e. it represents an electrical load. This is why we refer to load operation here. The positive reactive power indicates the prevailing effects of a coil and thus has an inductive effect. The apparent power in the first quadrant is operated in inductive load mode. If a negative active power affects the apparent power, active power is delivered, as in the case of a generator that delivers power. With positive reactive power, inductive generator operation occurs (quadrant II). If the reactive power is negative, capacitive effects predominate. With negative active power, this type of operation is called capacitive generator operation (quadrant III). Finally, capacitive load operation is described in the IV. quadrant. Here, positive active power and negative reactive power affect the apparent power.

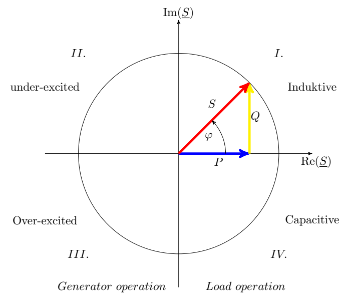


Figure 6.4: **Pointer diagram of the Power.** Composition of reactive, active and apparent power depending on the phase shift between current and voltage. Here, an inductive load in the positive imaginary part.

Reactive power is needed, among other things, to generate magnetic fields in electrical machines. The reactive power requirement arises from components in the system that have either an inductive or capacitive effect. If the current required to generate the magnetic field (excitation current) is higher than would be necessary for the nominal power of the generator, the generator is overexcited. In this case, the generator is able to supply reactive power to the grid. The resulting increase in reactive power in the grid leads to a rise in voltage. If the energy demand in the grid increases, this situation can be used to counteract a drop in grid voltage. Conversely, a generator with a lower excitation current than required operates in an underexcited state. The generator draws reactive power from the grid in order to maintain the required magnetic field. The consumption of reactive power then leads to a voltage drop in the grid. Severe underexcitation of a generator can cause the electrical machine to fail.

The units of the three types of power are theoretically all comparable. In order to distinguish between the types of power, new units were chosen for apparent power and reactive power, which physically express the same thing. The unit volt-ampere [VA] was defined for apparent power and volt-ampere reactive [var] for reactive power. Active power is specified in the familiar unit of watts [W].

Table 6.1: **Summary of power types.** Active power, reactive power and apparent power caused by alternating quantities, listed with their symbols and units.

Power type		**Unit**	
Active power	P	W	(Watt)
Reactive power	Q	var	(Volt-Ampere-reaktive)
Apparant power	S	VA	(Volt-Ampere)

6.4 Power and electrical energy

In addition to electrical power, electrical energy is also a quantity used, for example, to record consumption in households. Here, the electrical power per unit of time is measured. According to the definition from thermodynamics, energy is the power applied over a certain period of time. So, according to equation 6.20, the electrical power during a time between t_1 and t_2 is integrated to determine the electrical energy.

$$E_{\mathrm{el}} = \int_{t_1}^{t_2} p_{\mathrm{el}}(t)\ \mathrm{d}t \tag{6.20}$$

Similar to active power and reactive power, energy can also be divided into active work and reactive work. Reactive work describes the portion of electrical energy that is not converted into useful energy or active work.

$$W_{\mathrm{N}} = \int_{t_1}^{t_2} p(t)\ \mathrm{d}t \qquad\qquad W_{\mathrm{Q}} = \int_{t_1}^{t_2} Q\ \mathrm{d}t \tag{6.21}$$

In a steady state, only the RMS value of voltage and current, and thus also of apparent power, active power and reactive power, are considered and are therefore constant over time.

$$W_{\mathrm{N}} = P \cdot (t_2 - t_1) \qquad\qquad W_{\mathrm{Q}} = Q \cdot (t_2 - t_1) \tag{6.22}$$

Example 6.1: Power calculation

The following values for the alternating voltage and alternating current are measured on an alternating current load:

Given is:

$$\hat{U} = 12\ V \qquad \hat{I} = 2A \qquad \varphi = 60°$$

The following tasks are to be completed:

a) Calculation of active power and reactive power.

b) Calculation of apparent power using the conjugate complex current.

a) The active power is:

$$P = U \cdot I \cdot \cos(\varphi) = \frac{\hat{U}}{\sqrt{2}} \cdot \frac{\hat{I}}{\sqrt{2}} \cdot \cos(\varphi) = \frac{12\ V}{\sqrt{2}} \cdot \frac{2\ A}{\sqrt{2}} \cdot \cos(60^o)$$

$$P = 6\ W$$

The reactive power is:

$$Q = U \cdot I \cdot \sin(\varphi) = \frac{\hat{U}}{\sqrt{2}} \cdot \frac{\hat{I}}{\sqrt{2}} \cdot \sin(\varphi) = \frac{12\ V}{\sqrt{2}} \cdot \frac{2\ A}{\sqrt{2}} \cdot \sin(60^o)$$

$$Q = 10{,}392\ var$$

b) The apparent power is:

$$\underline{S} = \underline{U} \cdot \underline{I}^* = \frac{\underline{\hat{U}}}{\sqrt{2}} \cdot \frac{\underline{\hat{I}}^*}{\sqrt{2}} = \frac{12\ V}{\sqrt{2}} \cdot e^{j(2\pi \cdot 20\ Hz)} \cdot \frac{2\ A}{\sqrt{2}} \cdot e^{j(2\pi \cdot 20\ Hz + \frac{\pi}{3})}$$

$$\underline{S} = 12\ VA \cdot e^{j\frac{\pi}{3}} = 6\ W + j10{,}392\ var$$

7 Three-phase current

To ensure the most efficient energy transport possible, three-phase current is used in many grid systems. Three-phase current is based on alternating current, which is not taken as a single phase, but as a combination of several phases. Alternating current is generated by generators that have not just a single winding in the rotor, but several. If these windings are installed at a symmetrical distance from each other, phases are generated depending on the number of windings. This is then referred to as a multi-phase system. If there are exactly three phases, the system is called three-phase current. The basics of alternating current and complex calculations are particularly relevant for understanding three-phase current.

Learning objectives: Drehstrom

The students

- understand the basic characteristics of three-phase current and the associated operators.
- can analyse three-phase systems with the same load (symmetrical components).
- are familiar with the characteristics of three-phase systems with different phase loads (unbalanced components).

7.1 Symmetrical components

In the field of energy generation, transmission and distribution, three-phase current is mainly used. The term 'three-phase current' comes from the technical fact that the systems are connected to the spatially rotating magnetic field of the generator: the rotating field. All phases in a multi-phase system are generated at the same frequency, which in Europe is known to be 50 Hz. The generated phases are connected at the point of generation, i.e. the generator, and behind the consumers. In a three-phase system, star connection or delta connection are available for this purpose. In the symmetrical case, the generator generates as many phases as there are windings in the rotor. The windings are always installed at the same angle. This angle is assigned the Greek letter α:

$$\alpha = \frac{2\pi}{m} \tag{7.1}$$

In the formula, m stands for the number of phases generated in the system. In a three-phase system, there are three phases. This results in an α of 120°.

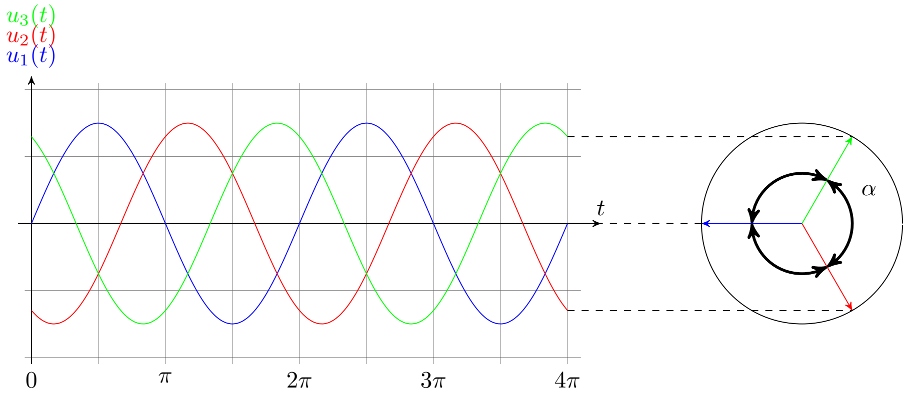


Figure 7.1: **Three phases of three-phase current.** The three phases of the three-phase system are all phase-shifted by 120° relative to each other. This results in an angle α of 120°.

The next step is to derive a rotation operator from the angle α. This rotation operator is used to convert the voltages and currents to the reference voltage/current.

$$\underline{a} = e^{j\alpha} = e^{j\frac{2\pi}{m}} \tag{7.2}$$

For $m = 3$:

$$\begin{aligned}
\underline{a} &= e^{j\frac{2\pi}{3}} = -\frac{1}{2} + j\frac{\sqrt{3}}{2} \\
\underline{a}^2 &= e^{j\frac{4\pi}{3}} = -\frac{1}{2} - j\frac{\sqrt{3}}{2} = \underline{a}^* \\
\underline{a}^3 &= e^{j2\pi} = 1 \\
\underline{a}^4 &= e^{j\frac{8\pi}{3}} = -\frac{1}{2} + j\frac{\sqrt{3}}{2} = \underline{a} \\
\underline{a}^5 &= e^{j\frac{10\pi}{3}} = -\frac{1}{2} - j\frac{\sqrt{3}}{2} = \underline{a}^2 \\
&\vdots
\end{aligned}$$

Key point: Rotation operator

The angle α indicates the phase shift between the phases. With the help of the rotation operator $\underline{a}$, voltages and currents can be converted.

The complex voltages of the respective outer conductors relative to a neutral conductor can be formulated using the following equation:

$$\underline{U}_{\mathrm{Lm}} = U_{\mathrm{L1}} \cdot e^{-j(m-1)\alpha} \tag{7.3}$$

The voltage U_1 is a real reference value for the multiphase system.

Now the equations for the rotation operator are combined with those for the voltage, resulting in the following relationship:

$$\begin{aligned}\underline{U}_{\mathrm{L1}} &= \underline{a}^m \cdot U_{\mathrm{L1}} = 1 \cdot U_{\mathrm{L1}} \\ \underline{U}_{\mathrm{L2}} &= \underline{a}^{m-1} \cdot U_{\mathrm{L1}} \\ &\vdots \\ \underline{U}_{\mathrm{Lm}-1} &= \underline{a}^2 \cdot U_{\mathrm{L1}} \\ \underline{U}_{\mathrm{Lm}} &= \underline{a} \cdot U_{\mathrm{L1}}\end{aligned}$$

With three phases, i.e. $m = 3$, the following relationship between the three voltages and the rotation operator results:

$$\underline{U}_{\mathrm{L1}} + \underline{U}_{\mathrm{L2}} + \underline{U}_{\mathrm{L3}} = U_1(1 + \underline{a}^2 + \underline{a}) \tag{7.4}$$

Note:

In the literature, the indices for the stresses are not always chosen uniformly. The most common alternative is the letters u, v, w for phases $1, 2, 3$. Other indices are also used, but these will not be discussed further here.

Fig. 7.2 illustrates once again how the rotation operators are composed in the complex numbers.

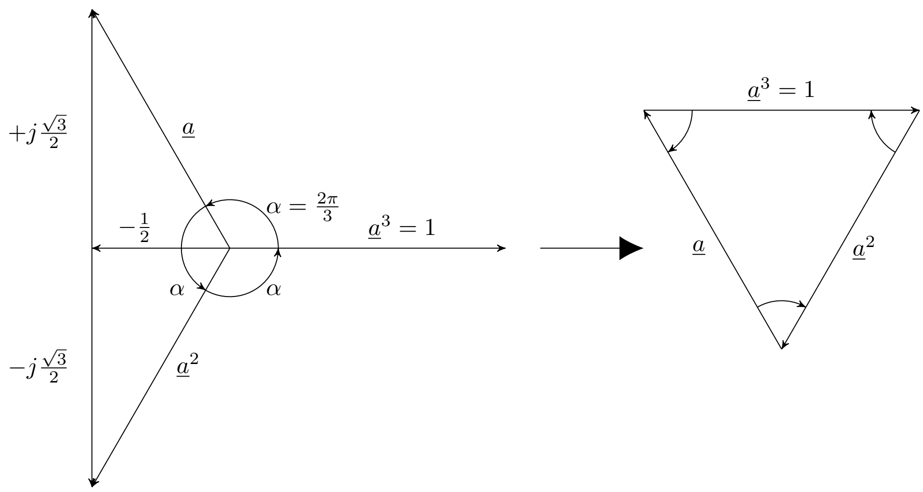


Figure 7.2: **Pointer diagram of the rotation operator in the three-phase system.** If the rotation operators are added together, the result is zero.

If you follow the path via the three rotation operators, you return to the origin and obtain zero.

$$\begin{aligned}1 + \underline{a} + \underline{a}^2 &= 0 \\ \underline{U}_{\mathrm{L1}} + \underline{U}_{\mathrm{L2}} + \underline{U}_{\mathrm{L3}} &= 0\end{aligned}$$

7.1.1 Line quantities in a star (wye) connection

The following chapters will discuss the types of connections that can occur in a three-phase system. First, we will look at what is known as a star connection. A star connection means that the phases are connected in such a way that a so-called star point is formed. A system can be connected on both the generator and consumer sides. Fig. 7.3 shows the typical components of a star connection in a three-phase equivalent circuit diagram. A voltage source is entered for each phase generated by the generator. If the generator has a star connection, the individual voltages are combined in a node in the ESB. This node is designated by the letter N_Q. A strand emanates from each voltage source, which should reflect the respective phase. Due to physical conditions, the components connected downstream of the generator, i.e. primarily the line and the consumer, exhibit both active and reactive resistance. These are combined in the ESB to form one impedance per strand. The impedances of the strands are combined in the star connection, as are the voltage sources, to form the node N_V. In a system in which both the generator and consumer sides are connected in a star configuration, there may be a neutral conductor or return conductor in addition to the phase strands, which connects the two nodes.

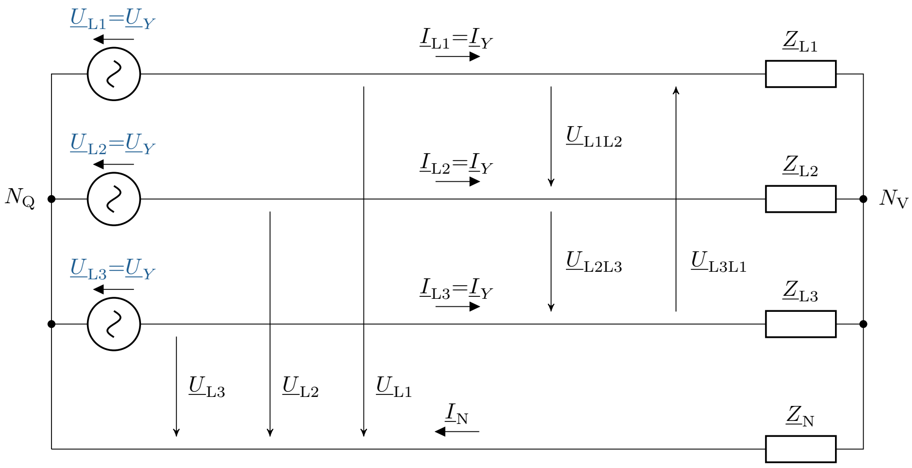


Figure 7.3: **ESB for a three-phase system.** Generators and consumers in the star and a neutral conductor between the nodes.

Voltages

In principle, there are three voltages in a three-phase system that should be specifically mentioned:

- Phase-to-neutral voltage (star voltage) U_Y: Voltage of the conductor relative to the neutral conductor (U_{L1}, U_{L2}, U_{L3})
- External conductor voltage (chained voltage) U: Voltage between two conductors (U_{L1L2}, U_{L2L3}, U_{L3L1}) – this voltage is also referred to as delta voltage
- Phase voltage U_{str}: Voltage across the impedance of the conductor relative to the neutral conductor. Depending on the connection, the phase voltage is either equal to the star voltage or the delta voltage.

Currents

For currents in a star system, it is helpful to apply Kirchhoff's first law. The sum of the currents in the strands must be equal to the current in the neutral conductor:

$$\underline{I}_N = \underline{I}_{L1} + \underline{I}_{L2} + \underline{I}_{L3} \tag{7.5}$$

In relation to voltage and impedance, Ohm's law states that:

$$\underline{I}_{\mathrm{N}} = \frac{\underline{U}_{\mathrm{L1}}}{\underline{Z}_1} + \frac{\underline{U}_{\mathrm{L2}}}{\underline{Z}_2} + \frac{\underline{U}_{\mathrm{L3}}}{\underline{Z}_3} = \underline{U}_{\mathrm{1m}}\underline{Y}_1 + \underline{U}_{\mathrm{2m}}\underline{Y}_2 + \underline{U}_{\mathrm{3m}}\underline{Y}_3 \tag{7.6}$$

When referring to a symmetrical system, this applies not only to the generator but also to the impedances. Accordingly, the impedances of the three strands are all equal:

$$\underline{Z}_{\mathrm{L1}} = \underline{Z}_{\mathrm{L2}} = \underline{Z}_{\mathrm{L3}} = \underline{Z} \tag{7.7}$$

If this relationship is taken into account for a symmetrical system for equation 7.7, the impedance can be extracted. It is already known that the sum of the voltages in a symmetrical system is zero. It can therefore be proven that the sum of the currents must also be zero.

$$\underline{I}_{\mathrm{N}} = \frac{1}{\underline{Z}}(\underline{U}_{\mathrm{L1}} + \underline{U}_{\mathrm{L2}} + \underline{U}_{\mathrm{L3}}) = 0 \tag{7.8}$$

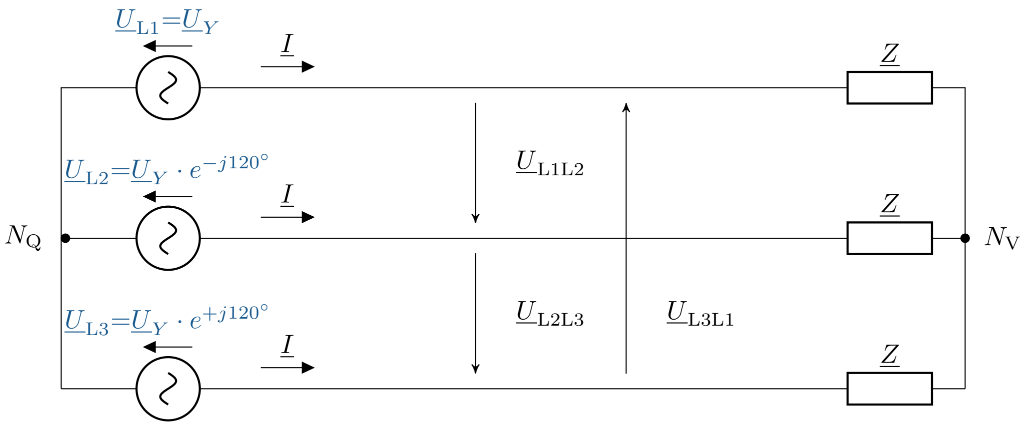


Figure 7.4: **Three-phase equivalent circuit diagram star-star.** ESB with symmetrical loads in star configuration

To avoid having to draw the complete three-phase ESB as shown in Fig. 7.4, it is possible to draw a simplified single-phase ESB. When drawing the single-phase ESB (and, of course, any other ESB), it is important to use the correct designations for the currents, voltages and impedances via the indices. The single-phase ESB for the star connection is shown in Fig. 7.5.

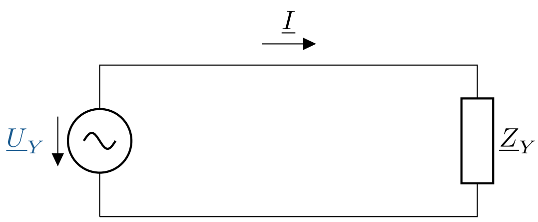


Figure 7.5: **Single-phase equivalent circuit diagram star-star.** ESB with symmetrical loads in star configuration

Key point: Symmetrical components

Symmetrical components describe an **equal load on the phases**, for example in a three-phase system. Here, all three phases are loaded by identical consumers.

7.1.2 Linked systems in a triangle

As an alternative to star connection, the sources and consumers can be connected in a delta configuration. However, in most cases, the generator side is connected in a star configuration. The reason for this is that even the smallest asymmetries cause compensating currents to flow, which would affect the operation of the generator. This should be avoided if possible, which is why a star connection is a better operating mode. For this reason, it is assumed that the generator side is fundamentally connected in a star configuration. So when we talk about a delta connection, unless explicitly stated otherwise, we are only referring to the consumer side.

It has already been defined that the outer conductor voltage is also called the delta voltage. According to Mesh Law, the delta voltage of two conductors is calculated from the difference between the respective star voltages:

$$\underline{U}_{\mathrm{L1L2}} = \underline{U}_{\mathrm{L1}} - \underline{U}_{\mathrm{L2}} \tag{7.9}$$

If the value of the respective outer conductor voltage is to be calculated, complex calculations would be required according to equation 7.9. However, a trigonometric derivation can be used to describe the relationship between star voltage and delta voltage.

$$\sin\left(\frac{\alpha}{2}\right) = \sin\left(\frac{\frac{2\pi}{m}}{2}\right) = \sin\left(\frac{\pi}{m}\right) = \frac{G}{H} \tag{7.10}$$

$$\begin{aligned} \sin\left(\frac{\pi}{3}\right) &= \frac{\frac{\underline{U}_{\mathrm{L1L2}}}{2}}{\underline{U}_{\mathrm{L1}}} \\ 2\cdot\frac{\sqrt{3}}{2} &= \frac{\underline{U}_{\mathrm{L1L2}}}{\underline{U}_{\mathrm{L1}}} \\ \underline{U}_{\mathrm{L1}} &= \frac{\underline{U}_{\mathrm{L1L2}}}{\sqrt{3}} \end{aligned} \tag{7.11}$$

Key point: Stern-Dreieck-Umrechnungsfaktor

The star voltage and the delta voltage can be converted into each other using the conversion factor $\sqrt{3}$:

$$U_Y = \frac{U_\Delta}{\sqrt{3}}$$

In Fig. 7.6, the phasor diagram of the complex rotating voltages is used to illustrate the relationship between star and delta voltages. For this purpose, trigonometric principles are used to represent the length of the outer conductor voltage (i.e. the delta voltage) in a different way. Equation 7.10 first describes in general terms how sine can be used in a multi-phase system. This equation applies in general, regardless of how many phases the system has. In equation 7.11, m is set to 3. This means that the angle is always half the actual angle between the phases. This means that the opposite side of the sine is half the delta voltage. By rearranging and converting the value $sin(\frac{\pi}{3})$ to $\frac{\sqrt{3}}{2}$, we finally obtain the factor needed to convert star and delta voltage.

In practice, star voltage is always marked with a star symbol as an index, while delta voltage usually does not require an index.

$$U_Y = \frac{U_\Delta}{\sqrt{3}} = \frac{U}{\sqrt{3}} \tag{7.12}$$

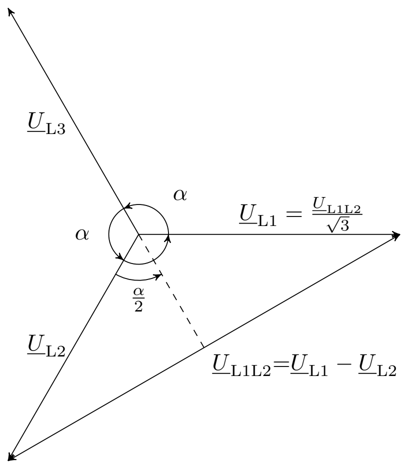


Figure 7.6: **Pointer diagram illustrating the derivation of the factor between star and delta connection.** The series voltage between two outer conductors $\underline{U}_{\mathrm{L1L2}}$ can be converted to the outer conductor voltage $\underline{U}_{\mathrm{L1}}$ using the conversion factor $\sqrt{3}$.

As can be seen in Figure 7.7, the impedances of the consumers are now connected in such a way that there is no longer a neutral point. In this circuit, the phase voltages now correspond to the outer conductor or delta voltage.

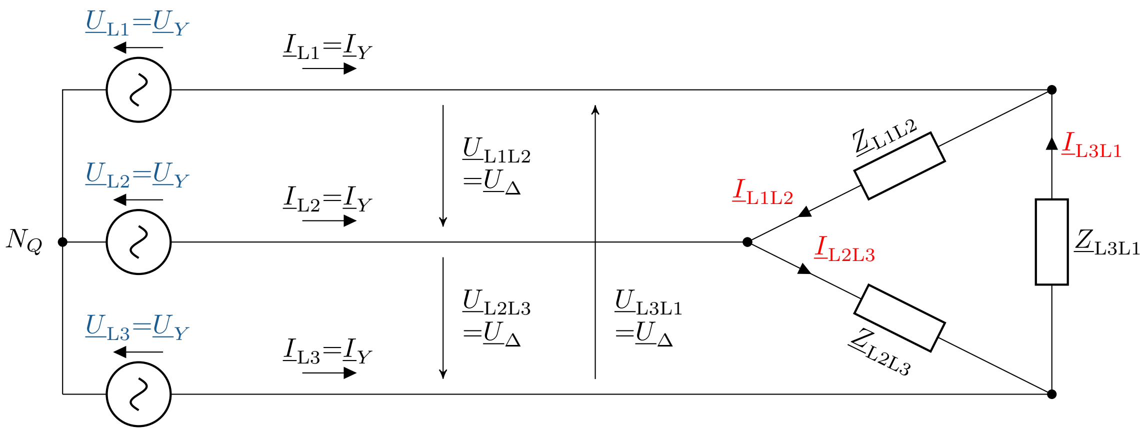


Abbildung 7.7: **Three-phase equivalent circuit diagram star-delta.** ESB with producers in the star and consumers in the triangle.

The currents between the strands can be determined from the node rule as follows:

$$\underline{I}_{\mathrm{L1}} = \underline{I}_{\mathrm{L1L2}} - \underline{I}_{\mathrm{L3L1}} \tag{7.13}$$
$$\underline{I}_{\mathrm{L2}} = \underline{I}_{\mathrm{L2L3}} - \underline{I}_{\mathrm{L1L2}} \tag{7.14}$$
$$\underline{I}_{\mathrm{L3}} = \underline{I}_{\mathrm{L3L1}} - \underline{I}_{\mathrm{L2L3}} \tag{7.15}$$

As with the star connection, in the symmetrical case it can be said that all impedances have the same value. This results in the following for the currents in the phase:

$$\underline{I}_{\mathrm{L1}} = \underline{I}_{\mathrm{L1L2}} - \underline{I}_{\mathrm{L3L1}} = \frac{\underline{U}_{\mathrm{L1L2}} - \underline{U}_{\mathrm{L3L1}}}{\underline{Z}} = \frac{U_Y}{\underline{Z}} \cdot (1 - \underline{a}^2 - \underline{a} + 1) = 3 \cdot \frac{U_Y}{\underline{Z}} \tag{7.16}$$
$$\underline{I}_{\mathrm{L2}} = \underline{I}_{\mathrm{L2L3}} - \underline{I}_{\mathrm{L1L2}} = \frac{\underline{U}_{\mathrm{L2L3}} - \underline{U}_{\mathrm{L1L2}}}{\underline{Z}} = \frac{U_Y}{\underline{Z}} \cdot (\underline{a}^2 - \underline{a} - 1 + \underline{a}^2) = 3 \cdot \underline{a}^2 \cdot \frac{U_Y}{\underline{Z}} \tag{7.17}$$
$$\underline{I}_{\mathrm{L3}} = \underline{I}_{\mathrm{L3L1}} - \underline{I}_{\mathrm{L2L3}} = \frac{\underline{U}_{\mathrm{L3L1}} - \underline{U}_{\mathrm{L2L3}}}{\underline{Z}} = \frac{U_Y}{\underline{Z}} \cdot (\underline{a} - 1 - \underline{a}^2 + \underline{a}) = 3 \cdot \underline{a} \cdot \frac{U_Y}{\underline{Z}} \tag{7.18}$$

These transformations show that the currents differ only in terms of the rotation operator a. Equivalent to the conversion from star to delta voltage, a ratio between phase and delta current can also be established. Using the same derivation as shown in Fig. 7.6, but replacing the voltage with the corresponding current values, the following ratio is obtained:

$$I_{\mathrm{str}} = \sqrt{3} \cdot I_{\Delta} \tag{7.19}$$

With the available information, a single-phase ESB can now also be set up for the delta connection (see Fig. 7.8).

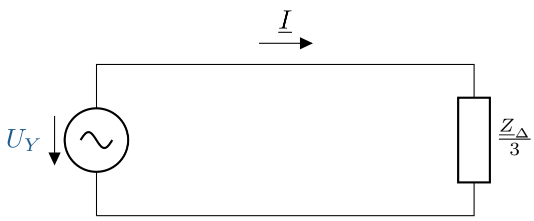


Figure 7.8: **Single-phase equivalent circuit diagram star-delta.** ESB of a delta connection with symmetrical consumers

7.1.3 Power in a symmetrical three-phase network

As in all electrical systems, power is calculated using the equation $P = U \cdot I$. In the case of a three-phase system, the power must be calculated per phase. In a symmetrical system, the currents and voltages per phase are equal, and the result for one phase must therefore be multiplied by 3.

$$\begin{aligned} S &= 3 \cdot I_{\mathrm{str}} \cdot U_Y = 3 \cdot I_{\mathrm{str}} \cdot \frac{U_{\Delta}}{\sqrt{3}} \\ &= 3 \cdot I_{\Delta} \cdot U_{\Delta} = 3 \cdot \frac{I_{\mathrm{str}}}{\sqrt{3}} \cdot U_{\Delta} \\ &= \sqrt{3} \cdot U_{\Delta} \cdot I_{\mathrm{str}} \end{aligned} \tag{7.20}$$

The formula 7.20 shows that it makes no difference to the power calculation which circuit is used. The only thing to bear in mind is which voltages and currents are used in the calculation (delta or star).

Note:

The notation of the indices for delta or star values is not clear in the literature. In most cases, no index is given for the delta voltage (external conductor voltage) and the phase current $U_\delta = U$ and $I_{ph} = I$, but only for the star voltage and the delta current.

If you directly compare the performance of a delta connection and a star connection, you get different results.

The following value is obtained for the star connection:

$$S = \sqrt{3} \cdot U \cdot I = \sqrt{3} \cdot U \cdot \frac{U_Y}{Z} = \frac{U^2}{Z} \tag{7.21}$$

For the delta connection, we again obtain:

$$S = \sqrt{3} \cdot U \cdot I = \sqrt{3} \cdot U \cdot \sqrt{3} \cdot \frac{U}{Z} = 3 \cdot \frac{U^2}{Z} \tag{7.22}$$

Accordingly, the power turnover at the same impedances in a delta connection is three times greater than in a star connection.

7.2 Unbalanced components

The fundamentals of a three-phase system were explained in the previous chapter. However, it was always assumed that both the generation and the consumers are symmetrical. Now let us consider how the system behaves when asymmetries are present. However, for the time being, the generator side remains symmetrical and connected in a star configuration. If the generation remains star-connected, there can be three different types of circuits:

- Four-wire network in star connection
- Three-wire network in star connection
- Three-wire network in delta connection

Key point: Unbalanced components

Unbalanced components unlike symmetrical components, they describe the **uneven load on the phases** due to different consumers.

7.2.1 Four-wire network in star connection

The structure of a four-wire network in star connection is shown as ESB in Fig. 7.3. The return conductor is connected to both star points, thus connecting the generator and consumer sides. Because the star voltages are symmetrical (based on the assumption that the generator side is symmetrical), the consumer voltages are also voltage-symmetrical. However, the currents differ from each other due to the different impedances. As a result, the current in the return conductor is usually not equal to zero.

$$\underline{I}_{\mathrm{N}} = \underline{I}_{\mathrm{L1}} + \underline{I}_{\mathrm{L2}} + \underline{I}_{\mathrm{L3}} \tag{7.23}$$

Logically, the power can then no longer be calculated as shown in formula 7.20. Each conductor must now be calculated individually and finally added together.

$$\underline{S} = \underline{I}^*_{\mathrm{L1}} \cdot \underline{U}_{\mathrm{L1}} + \underline{I}^*_{\mathrm{L2}} \cdot \underline{U}_{\mathrm{L2}} + \underline{I}^*_{\mathrm{L3}} \cdot \underline{U}_{\mathrm{L3}} \tag{7.24}$$

7.2.2 Three-wire network in star connection

In principle, the same conditions apply in a three-wire star connection as in a four-wire connection. The difference is that the current cannot flow back through the return conductor and therefore has an effect on the other conductors (see Fig. 7.9). The same applies to the voltage. Since there is no voltage drop in the return conductor, this affects the voltage across the loads. A voltage difference arises between the two star points, which must first be calculated using the voltages across the loads. Two methods can be used to determine the required values. For the first method, the impedances are converted mathematically into a delta connection so that they can be calculated easily using known methods. This is because in a delta connection, the voltages across the impedances are equal to the conductor voltages. The outer conductor currents can also be determined relatively easily.

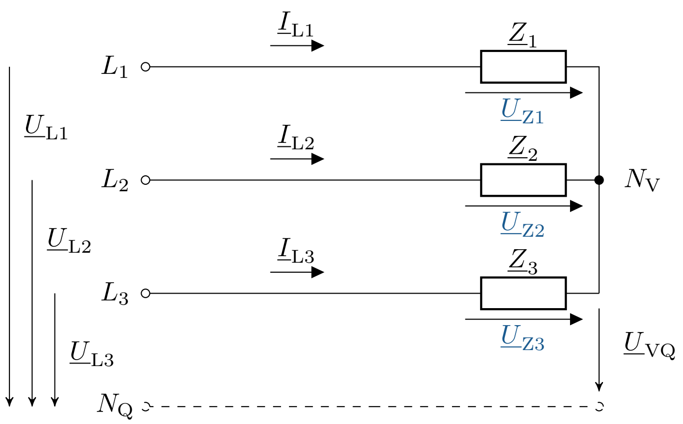


Figure 7.9: **Simplified ESB.** Three-wire network in star connection with unbalanced loads

With the alternative solution, equations can be set up using Ohm's and Kirchhoff's laws with the known values of the system. The following equation can be set up for the differential voltage between the two star points:

$$\underline{U}_{VQ} = \frac{\frac{\underline{U}_{\mathrm{L1}}}{\underline{Z}_1} + \frac{\underline{U}_{\mathrm{L2}}}{\underline{Z}_2} + \frac{\underline{U}_{\mathrm{L3}}}{\underline{Z}_3}}{\frac{1}{\underline{Z}_1} + \frac{1}{\underline{Z}_2} + \frac{1}{\underline{Z}_3}} = \frac{\underline{U}_{\mathrm{L1}} \cdot \underline{Z}_2 \cdot \underline{Z}_3 + \underline{U}_{\mathrm{L2}} \cdot \underline{Z}_3 \cdot \underline{Z}_1 + \underline{U}_{\mathrm{L3}} \cdot \underline{Z}_1 \cdot \underline{Z}_2}{\underline{Z}_1 \cdot \underline{Z}_2 + \underline{Z}_2 \cdot \underline{Z}_3 + \underline{Z}_3 \cdot \underline{Z}_1} \tag{7.25}$$

As is well known, the power is calculated from the sum of the powers of the individual phases. However, it is important to note that the calculation is not based on the outer conductor voltage, but on the voltages across the impedances:

$$\underline{S} = \underline{I}^*_{\mathrm{L1}} \cdot \underline{U}_{\mathrm{Z1}} + \underline{I}^*_{\mathrm{L2}} \cdot \underline{U}_{\mathrm{Z2}} + \underline{I}^*_{\mathrm{L3}} \cdot \underline{U}_{\mathrm{Z3}} \tag{7.26}$$

Theoretically, this formula could be used to perform the calculation. However, the measurements do not provide the values required for this formula. (How measurements are performed in a three-phase system will be explained in a later chapter.) If the formula for power is rearranged as follows, it can be seen that the sum of the conductor currents must be zero, and thus the last term is omitted.

$$\begin{aligned}\underline{S} &= \underline{I}_{\mathrm{L1}}^* \cdot \underline{U}_{\mathrm{L1}} + \underline{I}_{\mathrm{L2}}^* \cdot \underline{U}_{\mathrm{L2}} + \underline{I}_{\mathrm{L3}}^* \cdot \underline{U}_{\mathrm{L3}} - \underline{U}_{\mathrm{VQ}} \cdot (\underline{I}_{\mathrm{L1}}^* + \underline{I}_{\mathrm{L2}}^* + \underline{I}_{\mathrm{L3}}^*) \\ &= \underline{I}_{\mathrm{L1}}^* \cdot \underline{U}_{\mathrm{L1}} + \underline{I}_{\mathrm{L2}}^* \cdot \underline{U}_{\mathrm{L2}} + \underline{I}_{\mathrm{L3}}^* \cdot \underline{U}_{\mathrm{L3}} \end{aligned} \tag{7.27}$$

Here you can see that this formula for power is the same as in a four-wire network in star connection. However, a simplification can be made here, as the node rule can be used to express one current in terms of the other two currents:

$$\begin{aligned}\underline{S} &= \underline{I}_{\mathrm{L1}}^* \cdot \underline{U}_{\mathrm{L1}} + \underline{I}_{\mathrm{L2}}^* \cdot \underline{U}_{\mathrm{L2}} + (-\underline{I}_{\mathrm{L1}}^* - \underline{I}_{\mathrm{L2}}^*) \cdot \underline{U}_{\mathrm{L3}} \\ &= (\underline{U}_{\mathrm{L1}} - \underline{U}_{\mathrm{L2}}) \cdot \underline{I}_{\mathrm{L1}}^* + (\underline{U}_{\mathrm{L2}} - \underline{U}_{\mathrm{L3}}) \cdot \underline{I}_{\mathrm{L2}}^* \\ &= \underline{U}_{\mathrm{L3L1}} \cdot \underline{I}_{\mathrm{L1}}^* + \underline{U}_{\mathrm{L2L3}} \cdot \underline{I}_{\mathrm{L2}}^* \end{aligned} \tag{7.28}$$

7.2.3 Three-wire network in delta connection

In a delta connection, there is no neutral point, which is why the delta connection can only be implemented in a three-wire network. In the delta connection, the outer conductor voltages are also symmetrical due to the symmetrical feed. The currents adjust according to the impedances and can be calculated as already known:

$$\underline{I}_{\mathrm{L1}} = \underline{I}_{\mathrm{L1L2}} - \underline{I}_{\mathrm{L3L1}} \tag{7.29}$$

$$\underline{I}_{\mathrm{L2}} = \underline{I}_{\mathrm{L2L3}} - \underline{I}_{\mathrm{L1L2}} \tag{7.30}$$

$$\underline{I}_{\mathrm{L3}} = \underline{I}_{\mathrm{L3L1}} - \underline{I}_{\mathrm{L2L3}} \tag{7.31}$$

The sum of the outer conductor currents must, of course, always be zero, even with unbalanced currents. This results in the same calculation for the power calculation as for the star connection in a three-wire network and therefore also the same transformations.

$$\begin{aligned}\underline{S} &= \underline{I}_{\mathrm{L1L2}}^* \cdot \underline{U}_{\mathrm{L1L2}} + \underline{I}_{\mathrm{L2L3}}^* \cdot \underline{U}_{\mathrm{L2L3}} + \underline{I}_{\mathrm{L3L1}}^* \cdot \underline{U}_{\mathrm{L3L1}} \\ &= \underline{I}_{\mathrm{L1}}^* \cdot \underline{U}_{\mathrm{L1}} + \underline{I}_{\mathrm{L2}}^* \cdot \underline{U}_{\mathrm{L2}} + \underline{I}_{\mathrm{L3}}^* \cdot \underline{U}_{\mathrm{L3}} \end{aligned} \tag{7.32}$$

$$\begin{aligned}\underline{S} &= \underline{I}_{\mathrm{L1}}^* \cdot \underline{U}_{\mathrm{L1}} + \underline{I}_{\mathrm{L2}}^* \cdot \underline{U}_{\mathrm{L2}} + (-\underline{I}_{\mathrm{L1}}^* - \underline{I}_{\mathrm{L2}}^*) \cdot \underline{U}_{L3} \\ &= (\underline{U}_{\mathrm{L1}} - \underline{U}_{\mathrm{L2}}) \cdot \underline{I}_{\mathrm{L1}}^* + (\underline{U}_{\mathrm{L2}} - \underline{U}_{\mathrm{L3}}) \cdot \underline{I}_{\mathrm{L2}}^* \\ &= \underline{U}_{\mathrm{L3L1}} \cdot \underline{I}_{\mathrm{L1}}^* + \underline{U}_{\mathrm{L2L3}} \cdot \underline{I}_{\mathrm{L2}}^* \end{aligned} \tag{7.33}$$

8 Multiphase systems – positive system, negative system and zero system

The chapter on three-phase current explained how voltages, currents and power are composed in various three-phase systems. In the special case of symmetrical consumers, it was shown that a network calculation can be carried out relatively easily. The simplified representation using single-phase equivalent circuits is particularly helpful for the calculation. These simplifications cannot be applied to unbalanced loads, which means that even for small networks, more comprehensive network calculations become confusing and time-consuming. Imbalances occur, for example, due to network faults such as a single-pole short circuit, switching operations or unbalanced loads on the individual phases. The relationships between triggering events and the results are difficult to understand based on numerical values alone. In order to be able to calculate comprehensive and unbalanced networks, the known system can be transformed into three new systems. The following three systems are examined in more detail for this purpose:

Learning objectives: Multiphase systems

The students

- understand the functions of shunt, counter and zero systems.
- can calculate the three-phase power of the various systems.
- can create equivalent circuit diagrams for symmetrical sources, loads and lines.

8.1 Positive-, negative-, and zero-sequence systems

The fundamental goal of the transformation is to convert an asymmetrical system of n vectors into n systems with a symmetrical vector arrangement. For a three-phase system, this means that three systems must be created. Each conductor, i.e. L_1, L_2 and L_3, must be replaced by a system with a symmetrical vector arrangement. The transformation rule is based on the already known rotation operator $\underline{a}$ and $\underline{a}^2$ (see equation 7.2). The symmetry conditions for the current in the respective conductors can be established as follows (it should be noted here that everything that applies to the current also applies to the voltage!):

$$\begin{aligned} \underline{I}_{\mathrm{L1}} &= \underline{I}_0 + \underline{I}_1 + \underline{I}_2 \\ \underline{I}_{\mathrm{L2}} &= \underline{I}_0 + \underline{a}^2 \cdot \underline{I}_1 + \underline{a} \cdot \underline{I}_2 \\ \underline{I}_{\mathrm{L3}} &= \underline{I}_0 + \underline{a} \cdot \underline{I}_1 + \underline{a}^2 \cdot \underline{I}_2 \end{aligned} \tag{8.1}$$

The indices on the right-hand side of the equations represent the three systems: positive system (1), negative system (2) and zero system (0). If we look at the values of the positive system, the rotation operators result in clockwise phase currents. In the counter-system, the transformed phase currents are arranged in a left-rotating manner, and in the zero system, all components are aligned in the same direction. In principle, the zero system only occurs when the sum of all currents in the output system is not equal to zero.

In the next step, the so-called decomposition equations (equations 8.1) are rearranged to show the ratio of the current of the zero-sequence system to the conductor currents of the original system.

For the current in the zero-sequence system, the following applies by addition:

$$\begin{aligned}\underline{I}_{L1}+\underline{I}_{L2}+\underline{I}_{L3} &= 3\cdot\underline{I}_0+\underline{I}_1\cdot(1+\underline{a}^2+\underline{a})+\underline{I}_2\cdot(1+\underline{a}+\underline{a}^2)\\ &= 3\cdot\underline{I}_0 &(8.2)\\ \Leftrightarrow \underline{I}_0 &= \frac{1}{3}\cdot(\underline{I}_{L1}+\underline{I}_{L2}+\underline{I}_{L3}) &(8.3)\end{aligned}$$

To determine the current of the secondary system, the equations are first multiplied so that the pre-factors before the secondary current add up to 1. hen the decomposition equations are added together again.

$\underline{a}^3 = 1$; $\underline{a}^4 = \underline{a}$; $\underline{a}^5 = \underline{a}^2$; ...

$$\underline{I}_{L1} = \underline{I}_0+\underline{I}_1+\underline{I}_2 \tag{8.4}$$
$$\underline{a}\cdot\underline{I}_{L2} = \underline{a}\cdot\underline{I}_0+\underline{a}^3\cdot\underline{I}_1+\underline{a}^2\cdot\underline{I}_2 \tag{8.5}$$
$$\underline{a}^2\cdot\underline{I}_{L3} = \underline{a}^2\cdot\underline{I}_0+\underline{a}^3\cdot\underline{I}_1+\underline{a}\cdot\underline{I}_2 \tag{8.6}$$

$$\begin{aligned}\underline{I}_{L1}+\underline{a}\cdot\underline{I}_{L2}+\underline{a}^2\cdot\underline{I}_{L3} &= \underline{I}_0\cdot(1+\underline{a}+\underline{a}^2)+3\cdot\underline{I}_1+\underline{I}_2\cdot(1+\underline{a}^2+\underline{a}) &(8.7)\\ &= 3\cdot\underline{I}_1\\ \Leftrightarrow \underline{I}_1 &= \frac{1}{3}\cdot(\underline{I}_{L1}+\underline{a}\cdot\underline{I}_{L2}+\underline{a}^2\cdot\underline{I}_{L3})\end{aligned}$$

Finally, the current for the counter-system is calculated. The same principle is applied as for the system: first, the pre-factor is adjusted, then all decomposition equations are added:

$$\underline{I}_{L1} = \underline{I}_0+\underline{I}_1+\underline{I}_2 \tag{8.8}$$
$$\underline{a}^2\cdot\underline{I}_{L2} = \underline{a}^2\cdot\underline{I}_0+\underline{a}^4\cdot\underline{I}_1+\underline{a}^3\cdot\underline{I}_2 \tag{8.9}$$
$$\underline{a}\cdot\underline{I}_{L3} = \underline{a}\cdot\underline{I}_0+\underline{a}^2\cdot\underline{I}_1+\underline{a}^3\cdot\underline{I}_2 \tag{8.10}$$

$$\begin{aligned}\underline{I}_{L1}+\underline{a}^2\cdot\underline{I}_{L2}+\underline{a}\cdot\underline{I}_{L3} &= (1+\underline{a}^2+\underline{a})\underline{I}_0+(1+\underline{a}+\underline{a}^2)\cdot\underline{I}_1+3\cdot\underline{I}_2 &(8.11)\\ &= 3\cdot\underline{I}_2\\ \Leftrightarrow \underline{I}_2 &= \frac{1}{3}\cdot(\underline{I}_{L1}+\underline{a}^2\cdot\underline{I}_{L2}+\underline{a}\cdot\underline{I}_{L3})\end{aligned}$$

This results in the following equations for the currents of all systems:

$$\underline{I}_0 = \frac{1}{3}\cdot(\underline{I}_{L1}+\underline{I}_{L2}+\underline{I}_{L3}) \tag{8.12}$$
$$\underline{I}_1 = \frac{1}{3}\cdot(\underline{I}_{L1}+\underline{\mathrm{a}}\cdot\underline{I}_{L2}+\underline{a}^2\cdot\underline{I}_{L3}) \tag{8.13}$$
$$\underline{I}_2 = \frac{1}{3}\cdot(\underline{I}_{L1}+\underline{a}^2\cdot\underline{I}_{L2}+\underline{\mathrm{a}}\cdot\underline{I}_{L3}) \tag{8.14}$$

If you look closely at these three equations, you will see that conductor L1 does not have a rotation operator as a leading factor in any of the three equations. Therefore, conductor L1 should be selected as the reference conductor. This means that this is the conductor through which an imbalance occurs in the original system (e.g. due to a single-pole earth fault). The remaining symmetries in the original system can then be more easily considered via the other conductors Of course, any conductor can be taken as the reference conductor, since each conductor can be freely assigned an index (L1, L2, L3). However, this allows the initial requirement to be met: the symmetries of the real networks and equipment must be used to advantage!

As an example, an unbalanced current in a three-phase system and the transformation may look as follows:

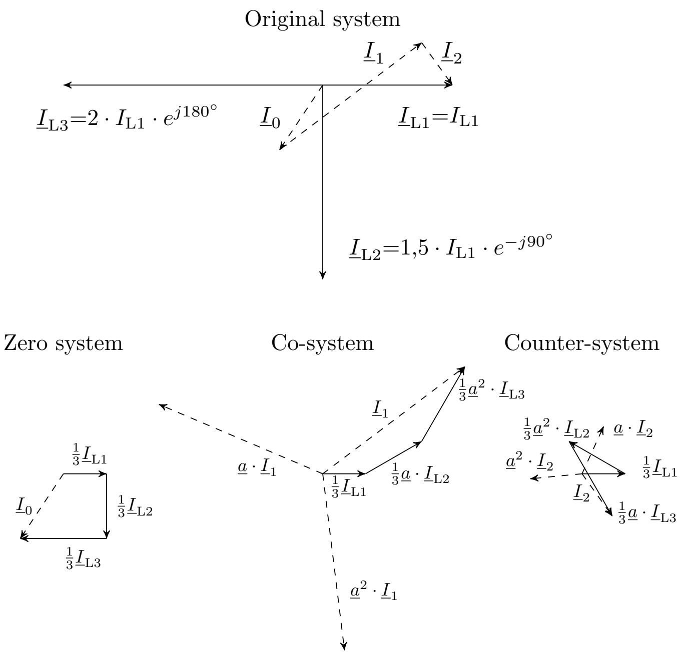


Figure 8.1: **Pointer diagrams of an unbalanced system.** Transformations from an original system into a positive, negative and zero system.

Fig. 8.1 shows the phasor diagram of an unbalanced load. In the original system, the phasors show that both the angles and the amplitudes deviate from the symmetrical state. The transformation into the three systems results in symmetrical phasors, which are easier to calculate with. The rotation operators and the prefactor of $\frac{1}{3}$ make it easy to see from the vectors how the transformed currents behave. To avoid having to write out the transformation equations in full every time, the equations can also be written in matrix notation:

$$\begin{bmatrix}\underline{I}_0\\ \underline{I}_1\\ \underline{I}_2\end{bmatrix} = \frac{1}{3}\cdot\begin{bmatrix}1 & 1 & 1\\ 1 & \underline{a} & \underline{a}^2\\ 1 & \underline{a}^2 & \underline{a}\end{bmatrix}\cdot\begin{bmatrix}\underline{I}_{L1}\\ \underline{I}_{L2}\\ \underline{I}_{L3}\end{bmatrix} \tag{8.15}$$

The notation can be simplified further by combining the term with the rotation operators and the term with the leading factor into a matrix. This matrix is also called the symmetrisation matrix:

$$[\underline{T}] = \frac{1}{3}\cdot\begin{bmatrix}1 & 1 & 1\\ 1 & \underline{a} & \underline{a}^2\\ 1 & \underline{a}^2 & \underline{a}\end{bmatrix} \tag{8.16}$$

In compact notation, this gives the following form:

$$\left[\underline{I}_{012}\right] = \left[\underline{T}\right] \cdot \left[\underline{I}_{\mathrm{L1L2L3}}\right] \tag{8.17}$$

According to the rules for calculating matrices, the inverse of $\left[\underline{T}\right]$ can be used to perform the inverse transformation:

$$\begin{bmatrix}\underline{I}_{\mathrm{L1}}\\ \underline{I}_{\mathrm{L2}}\\ \underline{I}_{\mathrm{L3}}\end{bmatrix} = \frac{1}{3} \cdot \begin{bmatrix}1 & 1 & 1\\ 1 & \underline{a}^2 & \underline{a}\\ 1 & \underline{a} & \underline{a}^2\end{bmatrix} \cdot \begin{bmatrix}\underline{I}_0\\ \underline{I}_1\\ \underline{I}_2\end{bmatrix} \tag{8.18}$$

$$\left[\underline{I}_{\mathrm{L1L2L3}}\right] = \left[\underline{T}\right]^{-1} \cdot \left[\underline{I}_{012}\right] \tag{8.19}$$

Key point: Positive-, negative-, and zero-sequence systems

A system is transformed into three systems for analysis: the partner system, the counter system, and the zero system.
The **Positive system** is assigned the index 1: $\underline{I}_1$
The **Negative system** is assigned the index 2: $\underline{I}_2$
The **Zero system** is assigned the index 0: $\underline{I}_0$

8.2 Three-phase power

How power is calculated has already been explained in the chapter on three-phase current. Here too, the equation for calculating power can be expressed in matrix notation for simplicity:

$$\begin{aligned}\underline{S}_{\mathrm{L1L2L3}} &= \underline{U}_{\mathrm{L1}} \cdot \underline{I}^*_{\mathrm{L1}} + \underline{U}_{\mathrm{L2}} \cdot \underline{I}^*_{\mathrm{L2}} + \underline{U}_{\mathrm{L3}} \cdot \underline{I}^*_{\mathrm{L3}} \\ &= \begin{bmatrix}\underline{U}_{\mathrm{L1}} & \underline{U}_{\mathrm{L2}} & \underline{U}_{\mathrm{L3}}\end{bmatrix} \cdot \begin{bmatrix}\underline{I}^*_{\mathrm{L1}}\\ \underline{I}^*_{\mathrm{L2}}\\ \underline{I}^*_{\mathrm{L3}}\end{bmatrix} = \left[\underline{U}_{\mathrm{L1L2L3}}\right] \cdot \left[\underline{I}_{\mathrm{L1L2L3}}\right]^*_t\end{aligned} \tag{8.20}$$

With the given rotation operators, the currents and voltages in a symmetrical system can be related to one phase each. ($\underline{U}_{\mathrm{L2}} = \underline{a}^2 \cdot \underline{U}_{\mathrm{L1}}, \underline{U}_{\mathrm{L3}} = \underline{a} \cdot \underline{U}_{\mathrm{L1}}$)

This results in the familiar simplification for power:

$$\underline{S}_{\mathrm{L1L2L3}} = \underline{U}_{\mathrm{L1}} \cdot \underline{I}^*_{\mathrm{L1}} + \underline{U}_{\mathrm{L1}} \cdot \underline{I}^*_{\mathrm{L1}} + \underline{U}_{\mathrm{L1}} \cdot \underline{I}^*_{\mathrm{L1}} \tag{8.21}$$

$$= 3 \cdot \underline{U}_{\mathrm{L1}} \cdot \underline{I}^*_{\mathrm{L1}} \tag{8.22}$$

Now, using matrix notation, the transformation into the new systems can be calculated:

$$
\begin{aligned}
[\underline{U}_{\text{L1L2L3}}] &= ([\underline{T}]^{-1} \cdot [\underline{U}_{012}]) = [\underline{U}_{012}] \cdot [\underline{T}]^{-1} && (8.23) \\
[\underline{I}_{\text{L1L2L3}}]^{*} &= ([\underline{T}]^{-1} \cdot [\underline{I}_{012}])^{*} = [\underline{T}]^{-1*} \cdot [\underline{I}_{012}]^{*} && (8.24) \\
\underline{S}_{\text{L1L2L3}} &= [\underline{U}_{012}]_{t} \cdot [\underline{T}]^{-1} \cdot [\underline{T}]^{-1*} \cdot [\underline{I}_{012}]^{*} && (8.25) \\
&= [\underline{U}_{012}]_{t} \cdot \begin{bmatrix} 1 & 1 & 1 \\ 1 & \underline{a}^2 & \underline{a} \\ 1 & \underline{a} & \underline{a}^2 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 & 1 \\ 1 & \underline{a}^2 & \underline{a} \\ 1 & \underline{a} & \underline{a}^2 \end{bmatrix}^{*} \cdot [\underline{I}_{012}]^{*} \\
&= [\underline{U}_{012}]_{t} \cdot \begin{bmatrix} 1 & 1 & 1 \\ 1 & \underline{a}^2 & \underline{a} \\ 1 & \underline{a} & \underline{a}^2 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 & 1 \\ 1 & \underline{a} & \underline{a}^2 \\ 1 & \underline{a}^2 & \underline{a} \end{bmatrix} \cdot [\underline{I}_{012}]^{*} \\
&= [\underline{U}_{012}] \cdot \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \cdot [\underline{I}_{012}]^{*} \\
&= 3 \cdot (\underline{U}_0 \cdot \underline{I}_o^{*} + \underline{U}_1 \cdot \underline{I}_1^{*} + \underline{U}_2 \cdot \underline{I}_2^{*}) = 3 \cdot \underline{S}_{012}
\end{aligned}
$$

8.3 Equivalent circuit diagram

ECDs fundamentally serve to simplify the observation and understanding of components. We will now examine various operating resources and how the transformations behave in different network situations.

8.3.1 Symmetrical voltage source

Symmetrical voltage sources can be found, for example, in mains supply and also in synchronous machines. The first step is to look at the output system. The relationships between the star voltage and the individual phases are already known from the chapter ???. Using the formula for the transformation (cf. 8.15), it can now be shown mathematically that in such a symmetrical case, the voltages of the counter and zero systems are 0.

Only in the neutral system does a driving voltage occur, equal to the star voltage:

$$
\begin{bmatrix} \underline{U}_0 \\ \underline{U}_1 \\ \underline{U}_2 \end{bmatrix} = \frac{1}{3} \cdot \begin{bmatrix} 1 & 1 & 1 \\ 1 & \underline{a} & \underline{a}^2 \\ 1 & \underline{a}^2 & \underline{a} \end{bmatrix} \cdot \begin{bmatrix} \underline{U}_{\text{L1}} \\ \underline{U}_{\text{L2}} \\ \underline{U}_{\text{L3}} \end{bmatrix} \tag{8.26}
$$

$$
[\underline{U}_{012}] = [\underline{T}] \cdot \begin{bmatrix} \underline{U}_Y \\ \underline{a}^2 \cdot \underline{U}_Y \\ \underline{a} \cdot \underline{U}_Y \end{bmatrix} \tag{8.27}
$$

Solving the equations for the transformed systems yields the following stress results:

$$
\underline{U}_0 = \frac{1}{3} \cdot U_Y \cdot (1 + \underline{a}^2 + \underline{a}) = 0 \tag{8.28}
$$

$$
\underline{U}_1 = \frac{1}{3} \cdot U_Y \cdot (1 + \underline{a} \cdot \underline{a}^2 + \underline{a}^2 \cdot \underline{a}) = U_Y \tag{8.29}
$$

$$
\underline{U}_2 = \frac{1}{3} \cdot U_Y \cdot (1 + \underline{a}^2 \cdot \underline{a}^2 + \underline{a} \cdot \underline{a}) = 0 \tag{8.30}
$$

With the results of the counter and zero systems, it can be said figuratively that these are short-circuited. Thus, the respective ECDs can be created for the three transformed systems:

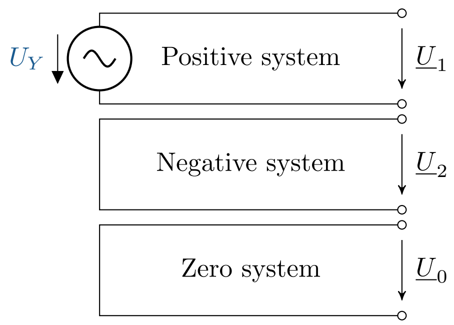


Figure 8.2: **Equivalent circuit diagrams of the transformed voltage source.** In the case of a symmetrical voltage source, only the ECD of the mitsystem has the voltage source.

In Figure 8.2, it should be clearly emphasised once again that, on the one hand, three individual systems are formed and, furthermore, that in the special case of symmetrical sources, a voltage only occurs in one of the systems.

8.3.2 Symmetrical load

A star connection with a return conductor should be selected for the consideration of the loads. An impedance Z is assumed in each conductor, including the return conductor. For the symmetrical case, the impedances are assumed to be equal; only the impedance in the return conductor may differ from the others and is given its own index.

For the original system, the equation for the voltage is set up as follows:

$$\underline{U}_{\mathrm{L1}} = \underline{Z} \cdot \underline{I}_{\mathrm{L1}} + \underline{Z}_{\mathrm{E}} \cdot (\underline{I}_{\mathrm{L1}} + \underline{I}_{\mathrm{L2}} + \underline{I}_{\mathrm{L3}}) \tag{8.31}$$
$$\underline{U}_{\mathrm{L2}} = \underline{Z} \cdot \underline{I}_{\mathrm{L2}} + \underline{Z}_{\mathrm{E}} \cdot (\underline{I}_{\mathrm{L1}} + \underline{I}_{\mathrm{L2}} + \underline{I}_{\mathrm{L3}}) \tag{8.32}$$
$$\underline{U}_{\mathrm{L3}} = \underline{Z} \cdot \underline{I}_{\mathrm{L3}} + \underline{Z}_{\mathrm{E}} \cdot (\underline{I}_{\mathrm{L1}} + \underline{I}_{\mathrm{L2}} + \underline{I}_{\mathrm{L3}}) \tag{8.33}$$

Here, too, matrix notation can be used to simplify the expression:

$$\begin{bmatrix} \underline{U}_{\mathrm{L1}} \\ \underline{U}_{\mathrm{L2}} \\ \underline{U}_{\mathrm{L3}} \end{bmatrix} = \begin{bmatrix} \underline{Z} + \underline{Z}_{\mathrm{E}} & \underline{Z}_{\mathrm{E}} & \underline{Z}_{\mathrm{E}} \\ \underline{Z}_{\mathrm{E}} & \underline{Z} + \underline{Z}_{\mathrm{E}} & \underline{Z}_{\mathrm{E}} \\ \underline{Z}_{\mathrm{E}} & \underline{Z}_{\mathrm{E}} & \underline{Z} + \underline{Z}_{\mathrm{E}} \end{bmatrix} \cdot \begin{bmatrix} \underline{I}_{\mathrm{L1}} \\ \underline{I}_{\mathrm{L2}} \\ \underline{I}_{\mathrm{L3}} \end{bmatrix} \tag{8.34}$$

With...

$$\left[\underline{Z}_{\mathrm{L1L2L3}}\right] = \begin{bmatrix} \underline{Z} + \underline{Z}_{\mathrm{E}} & \underline{Z}_{\mathrm{E}} & \underline{Z}_{\mathrm{E}} \\ \underline{Z}_{\mathrm{E}} & \underline{Z} + \underline{Z}_{\mathrm{E}} & \underline{Z}_{\mathrm{E}} \\ \underline{Z}_{\mathrm{E}} & \underline{Z}_{\mathrm{E}} & \underline{Z} + \underline{Z}_{\mathrm{E}} \end{bmatrix} \tag{8.35}$$

...results in

$$\left[\underline{U}_{\mathrm{L1L2L3}}\right] = \left[\underline{Z}_{\mathrm{L1L2L3}}\right] \cdot \left[\underline{I}_{\mathrm{L1L2L3}}\right] \tag{8.36}$$

If transformed using the transformation equations, the following applies:

$$\left[\underline{T}\right]^{-1} \cdot \left[\underline{U}_{012}\right] = \left[\underline{Z}_{\mathrm{L1L2L3}}\right] \left[\underline{T}\right]^{-1} \cdot \left[\underline{I}_{012}\right] \tag{8.37}$$

For symmetrical voltage sources, the voltage could be calculated from the transformation depending on the original system. The same should be done for the impedances for the consumers. To do this, the equations must be transformed in such a way that a ratio of transformed impedances and impedances of the original system is obtained. To achieve this, both sides are first expanded with $\left[\underline{T}\right]$ and then rearranged based on Ohm's law.

$$\left[\underline{T}\right] \cdot \left[\underline{T}\right]^{-1} \cdot \left[\underline{U}_{012}\right] = \left[\underline{T}\right] \cdot \left[\underline{Z}_{\mathrm{L1L2L3}}\right] \left[\underline{T}\right]^{-1} \cdot \left[\underline{I}_{012}\right] \tag{8.38}$$

$$\left[\underline{U}_{012}\right] = \left[\underline{T}\right] \cdot \left[\underline{Z}_{\mathrm{L1L2L3}}\right] \left[\underline{T}\right]^{-1} \cdot \left[\underline{I}_{012}\right] \tag{8.39}$$

$$\left[\underline{Z}_{012}\right] = \left[\underline{T}\right] \cdot \left[\underline{Z}_{\mathrm{L1L2L3}}\right] \left[\underline{T}\right]^{-1} \tag{8.40}$$

Performing the matrix operation of the last equation yields the desired relationship between the original and transformation systems.

$$\left[\underline{Z}_{012}\right] = \begin{bmatrix} \underline{Z} + 3\underline{Z}_E & 0 & 0 \\ 0 & \underline{Z} & 0 \\ 0 & 0 & \underline{Z} \end{bmatrix} \tag{8.41}$$

Now, the respective ECDs of the transformed systems can also be established for the symmetrical loads:

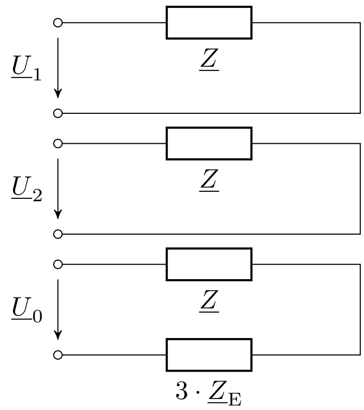


Figure 8.3: **Equivalent circuit diagrams of the transformed load.** With a symmetrical load, the three systems all have a symmetrical impedance, but the zero system also has three times the impedance for a return conductor.

In Fig. 8.3, the same impedances occur in the positive and negative systems. If the consumers are connected in a delta configuration, the impedances would have to be converted to star values using the known ratios. For zero impedances, currents only flow if there is actually a connection to a star. Then the impedances in the return conductor occur with triple the value.

8.3.3 Symmetrical lines

The line is the connection between the voltage source and the consumers. Until now, considerations regarding three-phase current have only distinguished between generation and consumers. With the lines, an additional component is now added that must be considered separately. Lines generally have their own impedances, coupling impedances and conductor-earth impedances. However, the conductor-earth impedances can be neglected. Each conductor has an intrinsic impedance and two coupling impedances, i.e. to the other conductors. The voltage for this calculation is defined as the difference between the voltages at the beginning and end of the line. This allows the matrix equation for a line system to be set up:

$$\begin{bmatrix} \underline{U}_{\mathrm{L1A}} - \underline{U}_{\mathrm{L1B}} \\ \underline{U}_{\mathrm{L2A}} - \underline{U}_{\mathrm{L2B}} \\ \underline{U}_{\mathrm{L3A}} - \underline{U}_{\mathrm{L3B}} \end{bmatrix} = \begin{bmatrix} \underline{Z}_{\mathrm{S}} & \underline{Z}_{\mathrm{K}} & \underline{Z}_{\mathrm{K}} \\ \underline{Z}_{\mathrm{K}} & \underline{Z}_{\mathrm{S}} & \underline{Z}_{\mathrm{K}} \\ \underline{Z}_{\mathrm{K}} & \underline{Z}_{\mathrm{K}} & \underline{Z}_{\mathrm{S}} \end{bmatrix} \cdot \begin{bmatrix} \underline{I}_{\mathrm{L1}} \\ \underline{I}_{\mathrm{L2}} \\ \underline{I}_{\mathrm{L3}} \end{bmatrix} \tag{8.42}$$

From this, a compact notation can be derived (with $U_{\mathrm{L1A}} - U_{\mathrm{L1B}} = U_{\mathrm{L1}}$, ...):

$$\left[\underline{U}_{\mathrm{L1L2L3}}\right] = \left[\underline{Z}_{\mathrm{L1L2L3}}\right] \cdot \left[\underline{I}_{\mathrm{L1L2L3}}\right] \tag{8.43}$$

We now have an equation that is very similar to the one for symmetrical consumers. Here, too, the equation must be transformed so that there is a direct dependency between the original system and the transformation system. Since the equation has the same basic structure, the same extensions and transformations can be carried out as for symmetrical consumers:

$$\left[\underline{Z}_{012}\right] = \begin{bmatrix} \underline{Z}_{\mathrm{S}} + 2 \cdot \underline{Z}_{\mathrm{K}} & 0 & 0 \\ 0 & \underline{Z}_{\mathrm{S}} - \underline{Z}_{\mathrm{K}} & 0 \\ 0 & 0 & \underline{Z}_{\mathrm{S}} - \underline{Z}_{\mathrm{K}} \end{bmatrix} \tag{8.44}$$

When transforming, it is always important that the transformed impedance matrix consists only of values on the main diagonal so that there are only direct connections between the individual conductors and the transformation systems.

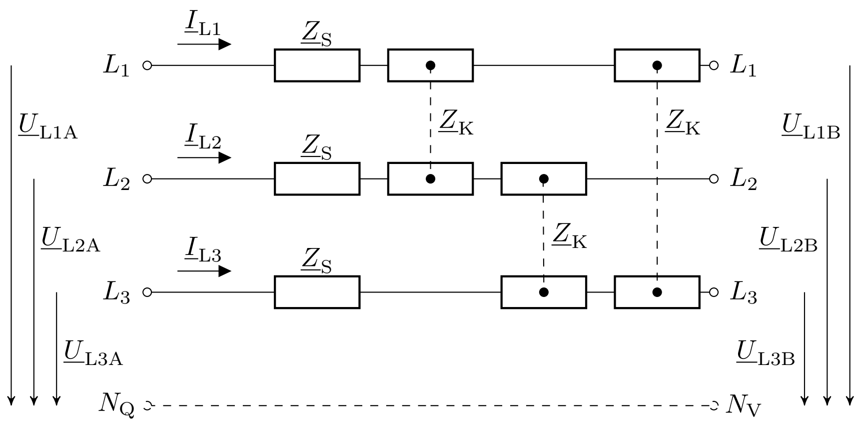


Figure 8.4: **Equivalent circuit diagrams of the transformed line.** ESB of a symmetrical line with the three phases L_1, L_2 and L_3 as well as the dotted line indicating the star point treatment.

Fig. 8.4 shows the ESB of the three systems for symmetrical lines. The dashed line is intended to reflect the neutral point treatment on the generation and consumption side for reference to the star voltages.

A Exercise tasks

A.1 Complex numbers 1

This applies to the following complex numbers:
$Z_1 = 3 + j4$
$Z_2 = 2 - j$
$Z_3 = j7$

The following parts of the task are to be calculated:

a) $Z_1 + Z_2$

b) $Z_1 - Z_3$

c) Polar form of Z_1, Z_2 and Z_3

d) $Z_1 \cdot Z_2$

e) $\frac{Z_1}{Z_3}$

f) Pointer diagram of $Z_1 + Z_2$ and $Z_1 \cdot Z_2$

A.2 Complex numbers 2

This applies to the following complex numbers:
$K_1 = 10 + j40$
$K_2 = 50 \cdot e^{-j40^o}$
$K_3 = 250 \cdot e^{j\frac{\pi}{4}}$

The following parts of the task are to be calculated:

a) $K_1 - K_2$, Result in polar coordinates

b) $K_3 + K_2$, result in Cartesian coordinates

c) $K_1 \cdot K_3$, result in polar coordinates

d) $K_1 + K_2$, graphically in the pointer diagram

e) $(K_1 - K_2)^2$

f) $\sqrt{K_1 + K_3}$

A.3 Pointer diagrams 1

Let the voltage be $\hat{U} = 325\ V \cdot \mathrm{e}^{\mathrm{j}30^\circ}$.

a) Draw the phasor diagram for the voltage $\hat{U}$ in the complex number space.

b) Calculate the real and imaginary parts of the complex voltage $\hat{U}$.

c) Explain the significance of the phase angle in relation to alternating voltage.

A.4 Complex AC calculation 1

The specific values for the resistance R, the coil L and the capacitor C are given.
$R = 10\ \Omega$, $L = 10\ mH$, $C = 10\ pF$
The impedances of the three components are to be calculated at the following frequencies:
$1\ \mu Hz$,$1\ mHz$,$1\ Hz$,$1\ kHz$,$1\ MHz$

Enter the results in a table.

A.5 Complex AC calculation 2

An impedance has a value of $\underline{Z} = (300 + \mathrm{j}400)\ \Omega$ at a frequency of $f = 1\ kHz$.

a) Determine the apparent resistance (magnitude) of the impedance.

b) Determine the admittance $\underline{Y} = \frac{1}{\underline{Z}}$ and the apparent conductance.

c) To realise the impedance $\underline{Z}$, specify a circuit consisting of two components connected in series and dimension this circuit.

d) To realise the impedance $\underline{Z}$, specify a circuit consisting of two components connected in parallel and dimension this circuit.

In addition, a voltage $u(t) = \hat{U} \cdot \cos(2\pi ft + \varphi_U)$ with $\hat{U} = 5\ V$, $\varphi_U = \pi/4$ and $f = 1\ kHz$ is applied to the impedance $\underline{Z} = (300 + \mathrm{j}400)\ \Omega$.

e) Determine the complex amplitude $\underline{\hat{U}}$ of the applied voltage.

f) Calculate the complex amplitude $\underline{\hat{I}}$ of the current through the impedance.

g) Specify the time dependence of the current $i(t)$.

h) Represent the impedance in polar form and specify the angle $\varphi = arg\{\underline{Z}\}$. Compare the angle of the impedance with the phase shift between voltage and current. What do you notice?

A.6 RMS value 1

Given a sinusoidal voltage

$$u(t) = 150\ V \cdot \sin(100\pi t)$$

where t is given in seconds and the voltage $u(t)$ is given in volts.

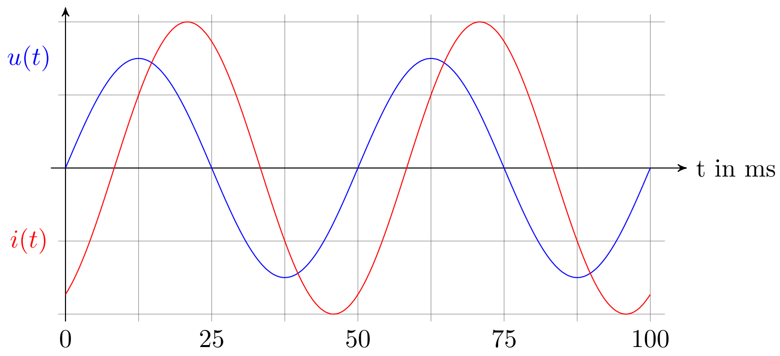
Calculate the following for this voltage:

a) The **arithmetic mean** $\bar{u}$ of the voltage over one period.

b) The **RMS value** U_{RMS} of the voltage.

A.7 Power calculation 1

Text



a) Reading the phase angle values of voltage and current. Describing the behaviour.

b) Specifying the period duration, frequency and angular frequency.

c) Calculate the instantaneous power at time t = 12.5 ms.

d) Calculate the active power and reactive power.

e) Calculate the apparent power using the conjugate complex current.

A.8 Three-phase current 1

A three-phase motor has the following nominal values:

- Nennspannung $U_N = 400\ V$
- Nennstrom $I_N = 10\ A$

a) Calculate the voltage across each winding of the motor in star connection $U_\star$.

b) Calculate the voltage across each winding of the motor in delta connection U_δ.

c) Calculate the current through each winding $I_\star$ in star connection.

d) Calculate the current through each winding I_Δ in delta operation.

A.9 Three-phase current 2

It is a symmetrical three-phase system with the following voltages:

$\underline{U}_1 = 230\ V$, $\underline{U}_2 = 230\ V \cdot \underline{a}^2$, $\underline{U}_3 = 230\ V \cdot \underline{a}$

The impedances all have a value of $Z = 10\Omega$

The following applies to all parts of the exercise: Solve the exercises with the least possible effort!

a) Draw the three-phase and single-phase ESBs in star and delta connections and label the ECDs correctly!

b) Calculate the magnitude of the consumer voltages for both the star and delta connections!

c) Calculate the complex consumer voltages in the delta connection!

d) Draw the phasor diagrams of the consumer voltages for the star and delta connections!

e) Calculate the complex load currents for the star connection and the magnitude of the load currents for the delta connection!

f) Calculate the apparent power for the star and delta connections!

A.10 Multiphase systems 1

Given is an unbalanced system with the following voltages:

$\underline{U}_{\mathrm{L1}} = 230V$, $\underline{U}_{\mathrm{L2}} = 460V \cdot e^{-j45^\circ}$, $\underline{U}_{\mathrm{L1}} = 345V \cdot e^{j120^\circ}$

a) Draw the phasor diagram of the original system!

b) Set up the decomposition equations for the voltages!

c) Calculate the voltages of the transformed systems!

d) Draw the phasor diagrams of the transformed systems!

B Solutions to the exercises

B.1 Complex numbers 1

a) $Z_1 + Z_2$

$$\begin{aligned} Z_1 + Z_2 &= (3 + j4) + (2 - j) \\ &= 5 + j3 \end{aligned}$$

b) $Z_1 - Z_3$

$$\begin{aligned} Z_1 - Z_3 &= (3 + j4) - (0 + j7) \\ &= 3 - j3 \end{aligned}$$

c) Polar form of Z_1, Z_2 and Z_3

$$\begin{aligned} Z_1 &= 5 \cdot \mathrm{e}^{\mathrm{j}53.13^\circ} \\ Z_2 &= 2{,}236 \cdot \mathrm{e}^{\mathrm{j}-26{,}57^\circ} \\ Z_3 &= 7 \cdot \mathrm{e}^{\mathrm{j}90^\circ} \end{aligned}$$

d) $Z_1 \cdot Z_2$

$$\begin{aligned} Z_1 \cdot Z_2 &= (3 + j4) \cdot (2 - j) \\ &= 10 + j5 \end{aligned}$$

e) $\frac{Z_1}{Z_3}$

$$\frac{Z_1}{Z_3} = \frac{3 + j4}{j7}$$

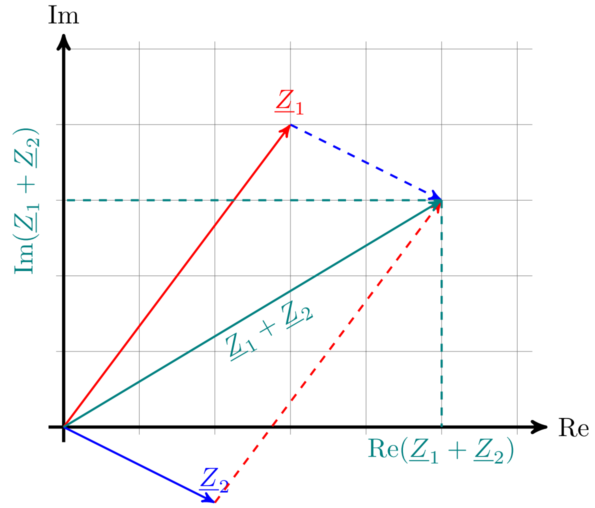
(Expand with the conjugate value of the denominator, i.e. -j7)

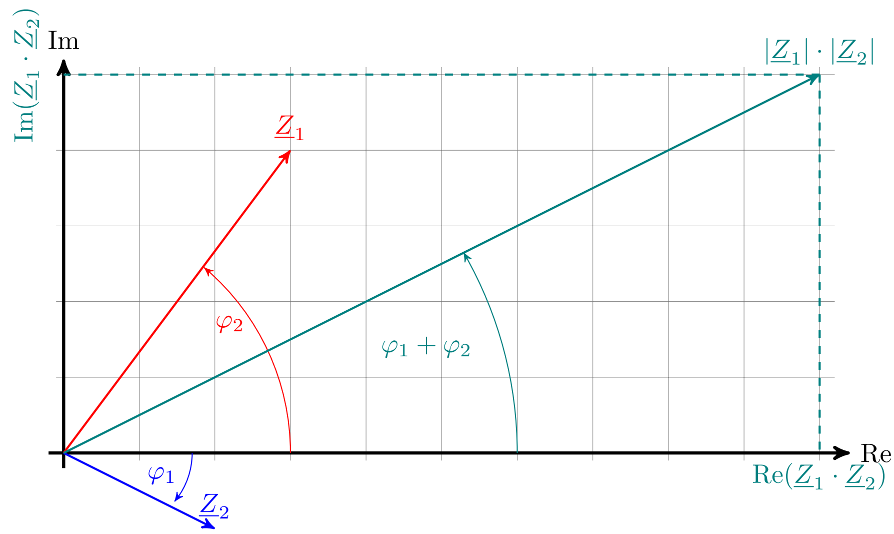
$$\frac{3 + j4}{j7} \cdot \frac{-j7}{-j7} = \frac{(3 + j4) \cdot (-j7)}{(j7) \cdot (-j7)}$$

Note: $j^2 = -1$

$$\begin{aligned} \frac{Z_1}{Z_3} &= \frac{28 - j21}{49} = \frac{28}{49} - j\frac{21}{49} = \frac{4}{7} - j\frac{3}{7} \\ &= \frac{4}{7} - j\frac{3}{7} \end{aligned}$$

f) Pointer diagram of $Z_1 + Z_2$ and $Z_1 \cdot Z_2$





B.2 Complex numbers 2

a) $K_1 - K_2$, Result in polar coordinates

$$10 + \mathrm{j} \cdot 40 - 50 \cdot \mathrm{e}^{-\mathrm{j}40^\circ} = ?$$

$\underline{K}_2$ convert to Cartesian coordinates:

$$\begin{aligned} 50 \cdot \mathrm{e}^{-\mathrm{j}40^\circ} &= 50 \cdot \cos(-40^\circ) + \mathrm{j}50 \cdot \sin(-40^\circ) \\ &= 50 \cdot 0{,}766 + \mathrm{j}50 \cdot (-0{,}6427) \\ &= 38{,}302 - \mathrm{j}32{,}1393 \end{aligned}$$

$\underline{K}_2 - \underline{K}_2$ Subtract in Cartesian coordinates:

$$\begin{aligned}
\underline{K}_1 - \underline{K}_2 &= (10 + \mathrm{j} \cdot 40) - (38{,}302 - \mathrm{j}32{,}1393) \\
&= 10 + \mathrm{j} \cdot 40 - 38{,}302 + \mathrm{j}32{,}1393 \\
&= -28{,}3 + \mathrm{j}72{,}139
\end{aligned}$$

Result in polar coordinates:

$$\begin{aligned}
\underline{K}_1 - \underline{K}_2 &= \sqrt{\mathrm{Re}^2 + \mathrm{Im}^2} \cdot \mathrm{e}^{\mathrm{j}\arctan(\frac{\mathrm{Im}}{\mathrm{Re}})} \\
&= 77{,}49 \cdot \mathrm{e}^{\mathrm{j}111{,}42^\circ}
\end{aligned}$$

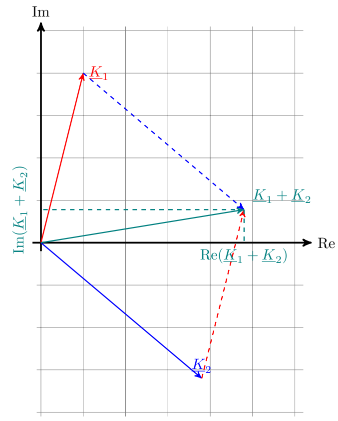
b) $K_3 + K_2$, Result in Cartesian coordinates

$$\begin{aligned}
K_3 &= 250 \cdot e^{j\frac{\pi}{4}} = 176{,}78 + j176{,}78 \\
K_2 &= 50 \cdot e^{-j40^\circ} = 38{,}3 - j32{,}15 \\
K_3 + K_2 &= (176{,}78 + j176{,}78) + (38{,}3 - j32{,}15) \\
K_3 + K_2 &= 215{,}08 + j144{,}63
\end{aligned}$$

c) $K_1 \cdot K_3$, Result in polar coordinates

$$\begin{aligned}
\underline{K}_1 &= 10 + \mathrm{j}40 \\
&= 41{,}23 \cdot \mathrm{e}^{\mathrm{j}75{,}96^\circ} \\
\underline{K}_1 \cdot \underline{K}_3 &= 10 + \mathrm{j}40 \cdot 250 \cdot \mathrm{e}^{\mathrm{j}45^\circ} \\
&= 10307{,}75 \cdot \mathrm{e}^{\mathrm{j}120{,}96^\circ}
\end{aligned}$$

d) $K_1 + K_2$, graphically in the pointer diagram



e) $(K_1 - K_2)^2$

Result from part a):

$$K_1 - Z_2 = 77{,}49 \cdot \mathrm{e}^{\mathrm{j}111{,}42^\circ}$$

Rule for the square of a complex number:

$$\begin{aligned}
(K_1 - K_2)^2 &= 77{,}49^2 \cdot \mathrm{e}^{\mathrm{j}2 \cdot 111{,}42^\circ} \\
&= 6004 \cdot \mathrm{e}^{-\mathrm{j}137^\circ}
\end{aligned}$$

f) $\sqrt{K_1 + K_3}$

$$K_1 + K_3 = 186{,}78 + j216{,}78$$

Conversion to polar form:

$$K_1 + K_3 = 286{,}09 \cdot \mathrm{e}^{\mathrm{j}49{,}32^\circ}$$

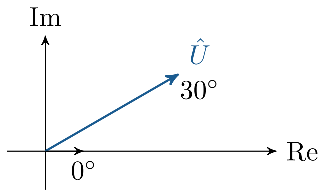
The square root of a complex number in polar form is:

$$\sqrt{r \cdot \mathrm{e}^{\mathrm{j}\varphi}} = \sqrt{r}\frac{\varphi}{2}$$

$$\sqrt{286{,}09} \cdot \mathrm{e}^{\mathrm{j}\frac{49{,}32^\circ}{2}} = 16{,}91 \cdot \mathrm{e}^{\mathrm{j}24{,}66^\circ}$$

B.3 Pointer diagrams 1

a) Pointer diagram for voltage $\hat{U}$



b) The real and imaginary parts of the voltage $\hat{U}$ can be calculated using the following formulas:

$$U_{\mathrm{Re}} = \hat{U} \cdot \cos(\varphi) = 325 \cdot \cos(30^\circ) \approx 281{,}46\ V$$
$$U_{\mathrm{Im}} = \hat{U} \cdot \sin(\varphi) = 325 \cdot \sin(30^\circ) = 162{,}5\ V$$

c) The phase angle φ describes the time lag between the voltage and the current in an alternating current circuit. A positive phase angle indicates that the voltage lags behind the current (inductive load), while a negative phase angle indicates that the current lags behind the voltage (capacitive load).

B.4 Complex AC calculation 1

Frequency	$\mathbf{X_R}$	$\mathbf{X_C}$	$\mathbf{X_L}$
$1\ \mu Hz$	$10\ \Omega$	$-j15{,}92\ M\Omega$	$j6{,}28$
$1\ mHz$	$10\ \Omega$	$-j15{,}92\ k\Omega$	k
$1\ Hz$	$10\ \Omega$	$-j15{,}92\ \Omega$	k
$1\ kHz$	$10\ \Omega$	$-j15{,}92\ m\Omega$	
$1\ MHz$	$10\ \Omega$	$-j15{,}92\ \mu\Omega$	

B.5 Complex AC calculation 2

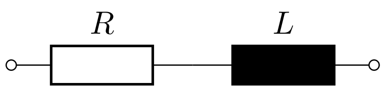
a)

$$
\begin{aligned}
|\underline{Z}| &= \sqrt{Re^2 + Im^2} \\
&= \sqrt{300^2 + 400^2}\ \Omega \\
&= 500\ \Omega
\end{aligned}
$$

b)

$$
\begin{aligned}
\underline{Y} &= \frac{1}{\underline{Z}} = \frac{1}{(300 + \mathrm{j}400)\ \Omega} \\
&= \frac{1}{(300 + \mathrm{j}400)\ \Omega} \cdot \frac{(300 - \mathrm{j}400)\ \cancel{\Omega}}{(300 - \mathrm{j}400)\ \cancel{\Omega}} \\
&= \frac{1}{300^2 + 400^2} \cdot (300 - \mathrm{j}400)\ S \\
&= (0.0012 - \mathrm{j}0.0016)\ S \\
|\underline{Y}| &= \frac{1}{|\underline{Z}|} \\
&= \frac{1}{500}\ S
\end{aligned}
$$

c) Series connection:
Real part $\rightarrow$ Ohmic resistance
Imaginary part $\rightarrow$ coil or capacitor?
Reactance coil: $X_L = 2\pi f \cdot L > 0$
Reactance capacitor: $X_C = -\frac{1}{2\pi f \cdot C} < 0$

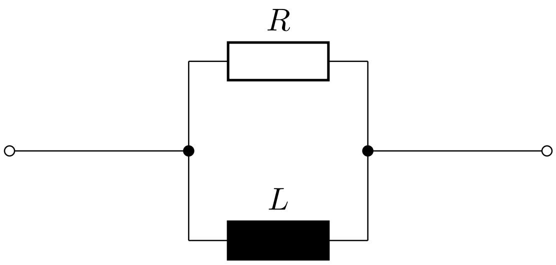


$$
\begin{aligned}
R &= 300\ \Omega \\
L &= \frac{X_L}{2\pi \cdot f} \\
&= \frac{400\ \Omega}{2\pi \cdot 1\ kHz} \\
&= 63.66\ mH \\
\underline{Z} &= R + \mathrm{j}X_L = R + \mathrm{j}\ 2\pi \cdot f \cdot L
\end{aligned}
$$

d) Parallel connection:
$\rightarrow$ Addition of guiding values:

$$
\underline{Y} = G + \mathrm{j}B = (0.0012 - \mathrm{j}0.0016)\ S \qquad \text{(aus b))}
$$

$B < 0 \rightarrow$ Coil, inductive reactance
$B > 0 \rightarrow$ Capacitor, capacitive reactance



$$\begin{aligned} R &= \frac{1}{G} \\ &= \frac{1}{0.0012\ S} \\ &= 833\ \Omega \\ B &= -\frac{1}{2\pi \cdot f \cdot L} \\ L &= \frac{1}{2\pi \cdot f \cdot B} \\ &= 99.47\ mH \end{aligned}$$

e)

$$\begin{aligned} \underline{U} &= \hat{U} \cdot \mathrm{e}^{\mathrm{j}\varphi_{\mathrm{U}}} = 5\ V \cdot \mathrm{e}^{\mathrm{j}\pi/4} = 5\ V \cdot \mathrm{e}^{\mathrm{j}45^\circ} \\ &= 5\ V \cdot (\cos(45^\circ) + \mathrm{j}\sin(45^\circ)) \\ &= 5\ V \cdot \frac{\sqrt{2}}{2}(1+\mathrm{j}) \\ &\approx (3{,}53 + \mathrm{j}3{,}53)\ V \end{aligned}$$

f)

g)

h)

B.6 RMS value 1

a) The arithmetic mean of a function u(t) over a period T is given by:

$$\bar{u} = \frac{1}{T}\int_0^T u(t)dt$$

Since $u(t)$ represents a sine function and the sine wave is symmetrical over one period, the positive and negative parts of the wave cancel each other out. Therefore, the arithmetic mean for each sine wave is always zero:

$$\bar{u} = 0$$

b) The RMS value of a sinusoidal voltage $u(t)$ is calculated using the following formula:

$$U_{Eff} = \frac{\hat{U}}{\sqrt{2}}$$

In this case, $\hat{U} = 150\ V$ is the amplitude of the voltage.

$$U_{Eff} = \frac{150\ V}{\sqrt{2}} \approx \frac{150\ V}{1.414} \approx 106.1\ V$$

B.7 Power calculation 1

a) The voltage has a phase angle of $\varphi_{\mathrm{u}} = 0$ and the current has a phase angle of $\varphi_{\mathrm{i}} = \pi/3$. The lagging of the current indicates an inductive behaviour.

b) The period T is 50 ms.
The frequency is:

$$f = \frac{1}{T} = \frac{1}{50\ ms} = 20\ Hz$$

The angular frequency is:

$$\omega = 2\pi \cdot f = 2\pi \cdot 20\ Hz = 125{,}66\ Hz$$

c) The instantaneous power at time t = 12.5 ms is:

$$p(t) = u(t) \cdot i(t) = 12\ V \cdot \sin(2\pi \cdot 20\ Hz \cdot 0{,}0125\ s) \cdot 2\ A \cdot \sin(2\pi \cdot 20\ Hz \cdot 0{,}0125\ s + \frac{\pi}{3})$$
$$p(t) = 12\ W$$

d) The active power is:

$$P = U \cdot I \cdot \cos(\varphi) = \frac{\hat{U}}{\sqrt{2}} \cdot \frac{\hat{I}}{\sqrt{2}} \cdot \cos(\varphi) = \frac{12\ V}{\sqrt{2}} \cdot \frac{2\ A}{\sqrt{2}} \cdot \cos(60^o)$$
$$P = 6\ W$$

The reactive power is:

$$Q = U \cdot I \cdot \sin(\varphi) = \frac{\hat{U}}{\sqrt{2}} \cdot \frac{\hat{I}}{\sqrt{2}} \cdot \sin(\varphi) = \frac{12\ V}{\sqrt{2}} \cdot \frac{2\ A}{\sqrt{2}} \cdot \sin(60^o)$$
$$Q = 10{,}392\ var$$

e) The apparent power is:

$$\underline{S} = \underline{U} \cdot \underline{I}^* = \frac{\underline{\hat{U}}}{\sqrt{2}} \cdot \frac{\underline{\hat{I}}^*}{\sqrt{2}} = \frac{12\ V}{\sqrt{2}} \cdot e^{j(2\pi \cdot 20\ Hz)} \cdot \frac{2\ A}{\sqrt{2}} \cdot e^{j(2\pi \cdot 20\ Hz + \frac{\pi}{3})}$$
$$\underline{S} = 12\ VA \cdot e^{j\frac{\pi}{3}} = 6\ W + j10{,}392\ var$$

B.8 Three-phase current 1

a) Voltage across each winding of the motor U_Y in star operation:

$$U_Y = \frac{U_N}{\sqrt{3}} = \frac{400\ V}{\sqrt{3}} \approx 230.94\ V$$

b) Voltage across each winding of the motor U_Δ in delta operation:

$$U_\Delta = U_N = 400\ V$$

c) Current through each winding I_Y in star mode:

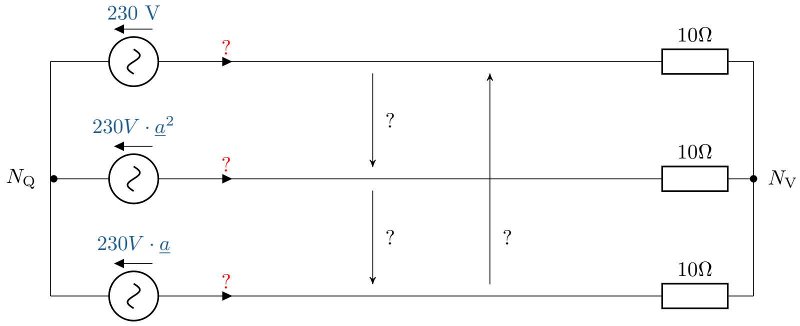
$$I_Y = I_N = 10\ A$$

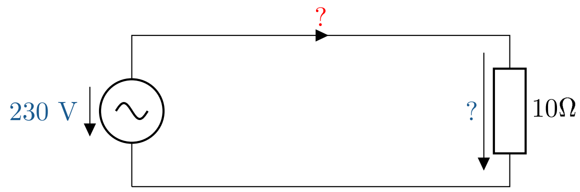
d) Current through each winding I_Δ in delta connection:

$$I_\Delta = \sqrt{3} \cdot I_Y = \sqrt{3} \cdot 10\ A \approx 17.32\ A$$

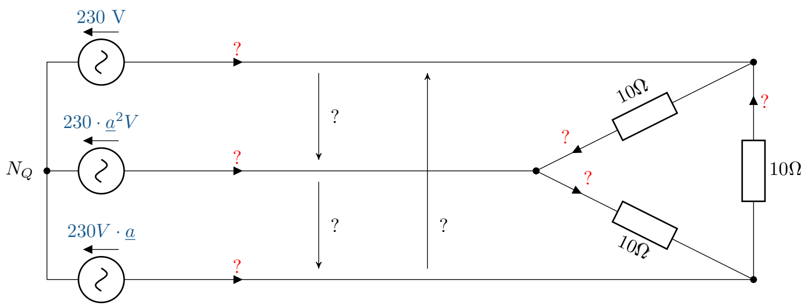
B.9 Three-phase current 2

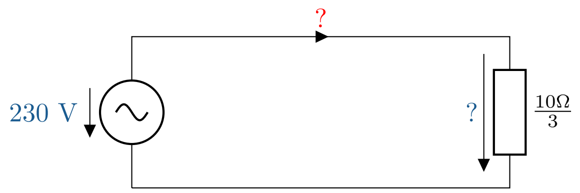
a) ECDs of the star connection:





ECDs of the triangular circuit:

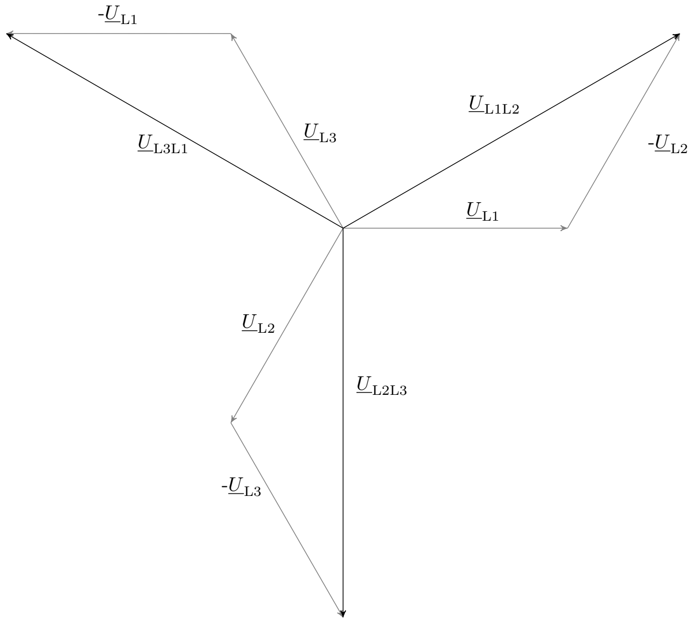




b) The amount of consumer voltage in a star connection is equal to the source voltage: 230 V. The amount of consumer voltage in a delta connection is $\sqrt{3}$ times greater than in a star connection: $230V \cdot \sqrt{3} = 398.37V \approx 400V$.

c) The complex consumer voltage in the delta connection is calculated from the differences in the star voltages.

$$\begin{aligned}
\underline{U}_{\mathrm{L1L2}} &= \underline{U}_{\mathrm{L1}} - \underline{U}_{\mathrm{L2}} = 230V - 230V \cdot \underline{a}^2 \\
&= 230V(1 - (-\frac{1}{2} - j\frac{\sqrt{3}}{2})) = \sqrt{3} \cdot 230V(\frac{3}{2} + j\frac{1}{2}) \\
&= \sqrt{3} \cdot 230V \cdot e^{j \cdot 30^\circ}
\end{aligned}$$



$$\begin{aligned}\underline{U}_{\mathrm{L2L3}} &= \underline{U}_{\mathrm{L2}} - \underline{U}_{\mathrm{L3}} = 230V \cdot \underline{a}^2 - 230V \cdot \underline{a} \\ &= 230V(-\frac{1}{2} - j\frac{\sqrt{3}}{2} - (-\frac{1}{2} + j\frac{\sqrt{3}}{2})) = \sqrt{3} \cdot 230V(-j) \\ &= \sqrt{3} \cdot 230V \cdot e^{-j \cdot 90^\circ}\end{aligned}$$

$$\begin{aligned}\underline{U}_{\mathrm{L3L1}} &= \underline{U}_{\mathrm{L3}} - \underline{U}_{\mathrm{L1}} = 230V \cdot \underline{a} - 230V \\ &= 230V(-\frac{1}{2} + j\frac{\sqrt{3}}{2} - 1) = \sqrt{3} \cdot 230V(-\frac{\sqrt{3}}{2} + j\frac{1}{2}) \\ &= \sqrt{3} \cdot 230V \cdot e^{j \cdot 150^\circ}\end{aligned}$$

d) Pointer diagrams of the consumer voltages for star and delta connections:

e) The current is calculated using Ohm's law. The following applies to star connections:

$$\begin{aligned}\underline{I}_1 &= \frac{\underline{U}_{\mathrm{L1}}}{\underline{Z}} = \frac{230V}{10\Omega} = 23A \\ \underline{I}_2 &= \frac{\underline{U}_{\mathrm{L2}}}{\underline{Z}} = \frac{230V \cdot \underline{a}^2}{10\Omega} = 23A \cdot \underline{a}^2 \\ \underline{I}_3 &= \frac{\underline{U}_{\mathrm{L3}}}{\underline{Z}} = \frac{230V \cdot \underline{a}}{10\Omega} = 23A \cdot \underline{a}\end{aligned}$$

With delta current, a distinction is made between the phase current and the delta current, i.e. the current through the consumers. The current in the consumer phases can be calculated using the voltage across the consumers in the delta circuit:

$$I_\Delta = \frac{U\Delta}{Z} = \frac{400V}{10\Omega} = 40A$$

The current in the strand is again $\sqrt{3}$ greater than in the consumer strands:

$$I_{\mathrm{str}} = \sqrt{3} \cdot I_\Delta = \sqrt{3} \cdot 40A = 69{,}28A$$

This value corresponds to three times the value of the current in a star connection.

f) The general power equation is as follows:

$$S = \sqrt{3} \cdot U_{\Delta} \cdot I_{\text{str}}$$

The current required depends on the circuit in question. As calculated in task 5, the phase current in the delta circuit is three times greater than in the star circuit.

The following applies to star connections:

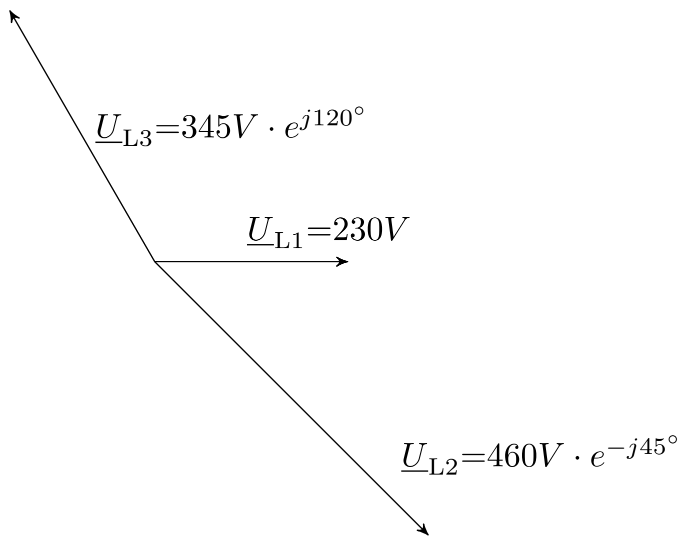
$$S = \sqrt{3} \cdot U \cdot I = \sqrt{3} \cdot 400V \cdot 23A = 15.934VA$$

And for the delta connection, the following applies:

$$S = \sqrt{3} \cdot U \cdot I = \sqrt{3} \cdot 400V \cdot 3 \cdot 23A = 47.804VA$$

B.10 Multiphase systems 1

a) Pointer diagram of the original system:



b)

$$\begin{aligned}
\underline{U}_{\text{L1}} &= 230V = \underline{U}_0 + \underline{U}_1 + \underline{U}_2 \\
\underline{U}_{\text{L2}} &= 460V \cdot e^{-j45°} = \underline{U}_0 + \underline{a}^2 \cdot \underline{U}_1 + \underline{a} \cdot \underline{U}_2 \\
\underline{U}_{\text{L3}} &= 345V \cdot e^{j120°} = \underline{U}_0 + \underline{a} \cdot \underline{U}_1 + \underline{a}^2 \cdot \underline{U}_2
\end{aligned}$$

c) For the zero system:

$$\begin{aligned}
\underline{U}_0 &= \frac{1}{3} \cdot (\underline{U}_{\text{L1}} + \underline{U}_{\text{L2}} + \underline{U}_{\text{L3}}) \\
&= \frac{1}{3} \cdot (230V + 460V \cdot e^{-j45°} + 345V \cdot e^{j120°}) \\
&= \frac{1}{3} \cdot (230V + 325{,}27V - j325{,}27V - 172{,}5V + 298{,}78V) \\
&= 127{,}59V - j8{,}83 = 127{,}9 \cdot e^{-j3{,}56°}
\end{aligned}$$

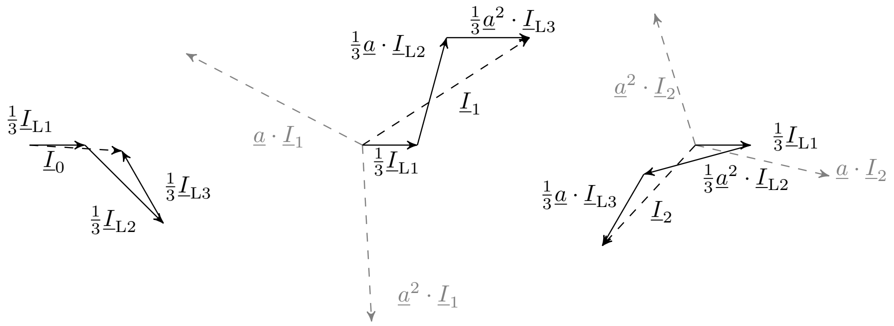
For the positive system:

$$
\begin{aligned}
\underline{U}_1 &= \frac{1}{3} \cdot (\underline{U}_{\mathrm{L1}} + \underline{a} \cdot \underline{U}_{\mathrm{L2}} + \underline{a}^2 \cdot \underline{U}_{\mathrm{L3}}) \\
&= \frac{1}{3} \cdot (230V + (-\frac{1}{2} + j\frac{\sqrt{3}}{2}) \cdot (325{,}27V - j325{,}7V) + (-\frac{1}{2} - j\frac{\sqrt{3}}{2}) \cdot (-172{,}5V + 298{,}78V)) \\
&= \frac{1}{3} \cdot (230V + 119{,}42V + j444{,}54V + 345V) \\
&= 231{,}48V + j148{,}18V = 274{,}84 \cdot e^{j32{,}63^\circ}
\end{aligned}
$$

For the negative system:

$$
\begin{aligned}
\underline{U}_2 &= \frac{1}{3} \cdot (\underline{U}_{\mathrm{L1}} + \underline{a}^2 \cdot \underline{U}_{\mathrm{L2}} + \underline{a} \cdot \underline{U}_{\mathrm{L3}}) \\
&= \frac{1}{3} \cdot (230V + (-\frac{1}{2} - j\frac{\sqrt{3}}{2}) \cdot (325{,}27V - j325{,}7V) + (-\frac{1}{2} + j\frac{\sqrt{3}}{2}) \cdot (-172{,}5V + 298{,}78V)) \\
&= \frac{1}{3} \cdot (230V + (-444{,}33V - j119{,}16V) + (-172{,}5V - 298{,}78V)) \\
&= -128{,}95V - j139{,}28V = 189{,}8 \cdot e^{j132{,}79^\circ}
\end{aligned}
$$

d) Pointer diagrams of the transformed systems:



Note: Creating the pointer diagrams is a good way to check the results calculated in step 3. The paths of the pointers taken from the original system must end where the pointer of the transformed system leads. If the end points do not match, an error has been made in one of the calculations.